

Installer's Guide

Air Handlers Convertible 2 – 5 Ton

Black Epoxy Coils

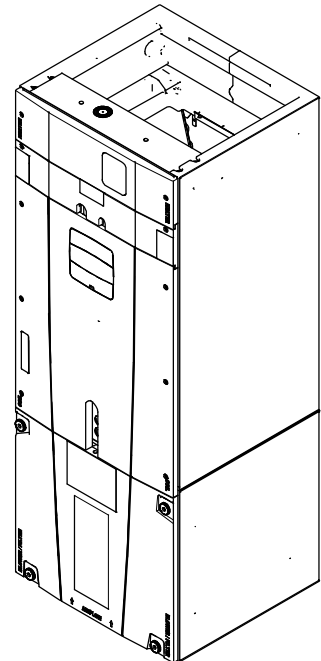
TAM7A0A24H21EA
TAM7A0B30H21EA
TAM7A0C36H31EA
TAM7A0C42H31EA
TAM7A0C48H41EA
TAM7B0C60H51EA

Standard Coils

TAM7A0A24H21SD
TAM7A0B30H21SD
TAM7A0C36H31SD
TAM7A0C42H31SD
TAM7A0C48H41SD
TAM7B0C60H51SC



Scan or click to view
[TAM7 Trouble Shooting
Video Library](#)



Note: "Graphics in this document are for representation only.
Actual model may differ in appearance."

Note: For use with BAYEV or BAYW series heaters ONLY

⚠ SAFETY WARNING

Only qualified personnel should install and service the equipment. The installation, starting up, and servicing of heating, ventilating, and air-conditioning equipment can be hazardous and requires specific knowledge and training. Improperly installed, adjusted or altered equipment by an unqualified person could result in death or serious injury. When working on the equipment, observe all precautions in the literature and on the tags, stickers, and labels that are attached to the equipment.

SAFETY SECTION

AIR HANDLERS

Important — This document contains a wiring diagram, a parts list, and service information. This is customer property and is to remain with this unit. Please return to service information pack upon completion of work.

⚠ WARNING

HAZARDOUS VOLTAGE!

Disconnect all electric power, including remote disconnects before servicing. Follow proper lockout/tagout procedures to ensure the power cannot be inadvertently energized.

⚠ CAUTION

GROUNDING REQUIRED!

Reconnect all grounding devices. All parts of this product that are capable of conducting electrical current are grounded. If grounding wires, screws, straps, clips, nuts, or washers used to complete a path to ground are removed for service, they must be returned to their original position and properly fastened.

⚠ WARNING

LIVE ELECTRICAL COMPONENTS!

Follow all electrical safety precautions when exposed to live electrical components. It may be necessary to work with live electrical components during installation, testing, servicing, and troubleshooting of this product.

⚠ WARNING

PRESSURIZED REFRIGERANT!

System contains oil and refrigerant under high pressure. Recover refrigerant to relieve pressure before opening the system. Do not use non-approved refrigerants or refrigerant substitutes or refrigerant additives.

⚠ CAUTION

SHARP EDGE HAZARD!

Be careful of sharp edges on equipment or any cuts made on sheet metal while installing or servicing.

Important: Panel damage can occur with prolonged exposure to POE lubricants. Air handler front panels that come in contact with POE oil must be washed immediately with soapy water.

Note: Representative illustrations only included in this document. Most illustrations display the upflow configuration.

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Installer Guide Notes

ALL Phases of this installation must comply with NATIONAL, STATE and LOCAL CODES!

Important: *This Document is customer property and is to remain with this unit. Please return to service information upon completion of work*

Important: *These instructions do not cover all variations in systems nor provide for every possible contingency to be met in connection with the installation. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to your installing dealer or local distributor.*

See TAM7 Service Facts document for information on reading the Control Board LEDs, Air Flow Tables and Troubleshooting Flowcharts.

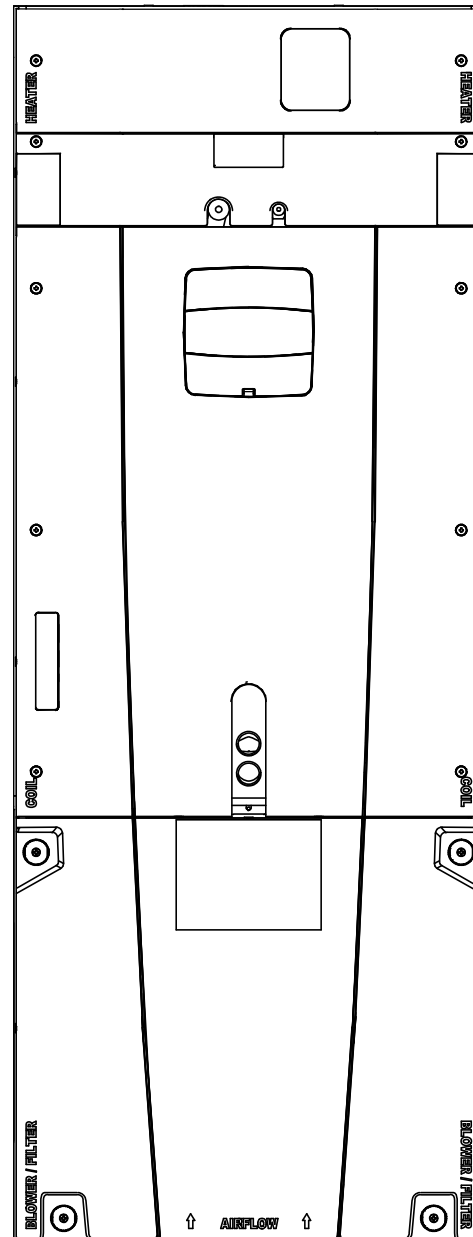
Note: *The manufacturer recommends installing ONLY A.H.R.I. approved, matched indoor and outdoor systems. Some of the benefits of installing approved matched indoor and outdoor split systems are maximum efficiency, optimum performance, and the best overall system reliability.*

Note: *Condensation may occur on the surface of the air handler when installed in unconditioned spaces, verify that all electrical and refrigerant line penetrations on the air handler are sealed completely.*

The TAM7 air handlers will only use the following internal electric heaters:

BAYEVAC04BK1	BAYEVAC10LG1
BAYEVAC04LG1	BAYEVC15BK1
BAYEVAC05BK1	BAYEVC20BK1
BAYEVAC05LG1	BAYEVCC25BK1
BAYEVAC08BK1	BAYEVAC10LG3
BAYEVAC08LG1	BAYEVC15LG3
BAYEVAC10BK1	

Note: *Duct heaters cannot be applied with this air handler.*



Unit Design

Table 1. Cabinet Penetration

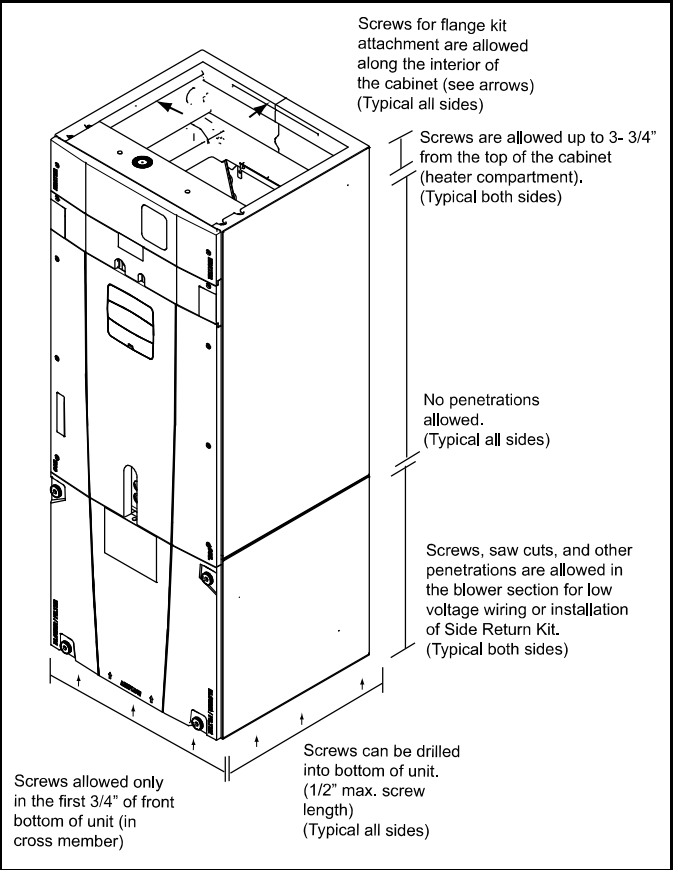
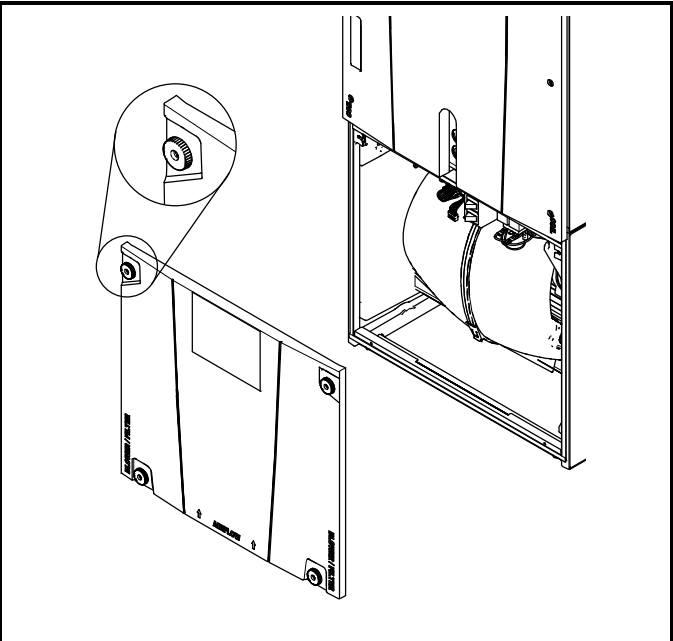
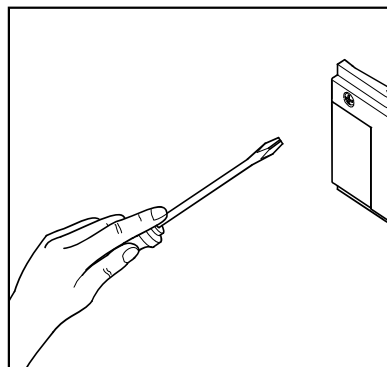
Important: Due to the unique design of this unit, which allows the electrical wiring to be routed within the insulation, do not screw, cut, or otherwise puncture the unit cabinet in any location other than the ones illustrated. Important: Under no conditions should metal strapping be attached to the unit to be used as support mechanisms for carrying or suspension purposes.	 Screws for flange kit attachment are allowed along the interior of the cabinet (see arrows) (Typical all sides) Screws are allowed up to 3- 3/4" from the top of the cabinet (heater compartment). (Typical both sides) No penetrations allowed. (Typical all sides) Screws, saw cuts, and other penetrations are allowed in the blower section for low voltage wiring or installation of Side Return Kit. (Typical both sides) Screws can be drilled into bottom of unit. (1/2" max. screw length) (Typical all sides) Screws allowed only in the first 3/4" of front bottom of unit (in cross member)
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Table 2. Panel Removal

The unit contains four (4) access panels: Blower/Filter, Coil, Line Set, and Heater. The Blower/Filter panel is removed using 1/4 turn thumb screws. 1. Turn thumb screws on Blower/Filter panel. 2. Pull top of panel out, away from cabinet. 3. Lift panel up out of channel. 4. Set aside.	
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The Coil, Line Set, and Heater panels are removed using Phillips head screws.

Removal requires #3 Size Phillips



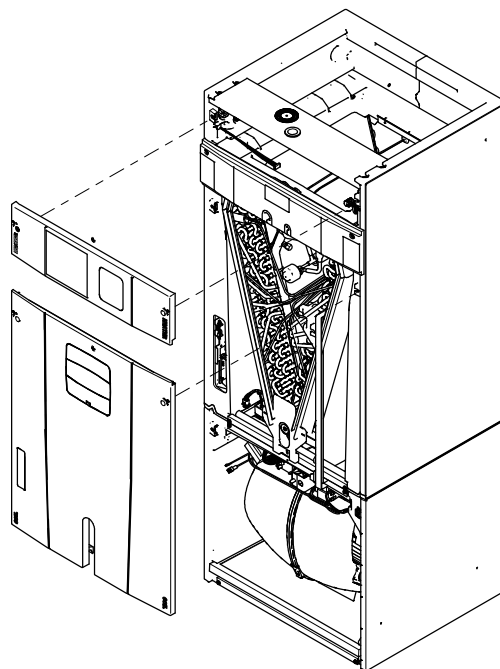
Coil and Heater panels must be removed prior to removing the Line Set panel.

To remove Coil Panel:

1. Turn screws on Coil panel.
2. Pull bottom of panel out, away from cabinet.
3. Pull panel down and out of channel.
4. Set aside.

To remove Heater Panel:

1. Turn screws on Heater panel.
2. Pull panel straight out, away from cabinet.
3. Set aside.

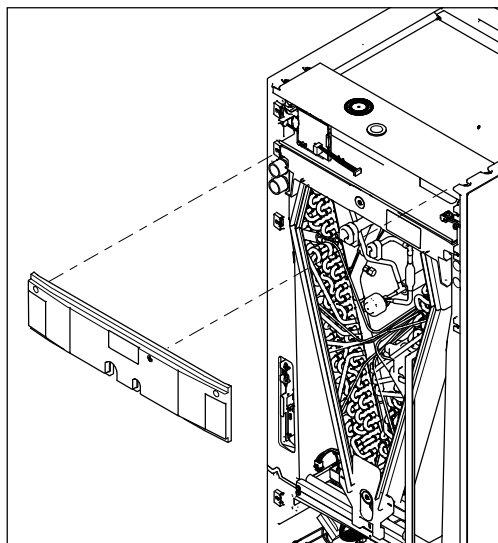


Removal of the Line Set panel is required for all refrigerant line brazing and some condensate line assembly depending on your orientation.

To remove Line Set panel:

1. Remove both Heater and Coil panels.
2. Turn screws on Line Set panel.
3. Pull panel straight out, away from cabinet.
4. Set aside.

Note: After replacing all panels, loosen the Line Set panel screws approximately 1/4 — 1/2 turn. This will improve the seal between the Heater Panel and Line Set panel.



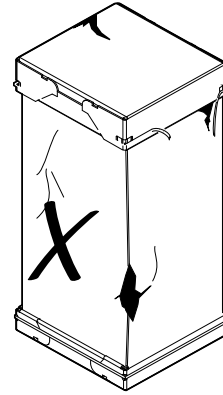
Unit Install Preparation

1. Check for damage and report promptly to the carrier any damage found to the unit.

Note: If the unit must be transported in a horizontal position, it must be laid on its back (marked "REAR" on carton).

Note: After unit is removed from the carton, verify coil is pressurized. Carefully remove the liquid line plug. If no pressure is released, check for leaks.

Note: Remove the cardboard from the bottom of the blower. Cut the tie wrap and remove the foam shipping block located at the motor.



Optional Equipment

Accessory Number	Description	Fits Cabinet Size
BAYEVAC04BK1	Electric Heater, 4kW, Breaker, RS-485 Control, 1 Ph	A to C
BAYEVAC04LG1	Electric Heater, 4kW, Lugs, RS-485 Control, 1 Ph	A to C
BAYEVAC05BK1	Electric Heater, 5kW, Breaker, RS-485 Control, 1 Ph	A to C
BAYEVAC05LG1	Electric Heater, 5kW, Lugs, RS-485 Control, 1 Ph	A to C
BAYEVAC08BK1	Electric Heater, 8kW, Breaker, RS-485 Control, 1 Ph	A to C
BAYEVAC08LG1	Electric Heater, 8kW, Lugs, RS-485 Control, 1 Ph	A to C
BAYEVAC10BK1	Electric Heater, 10kW, Breaker, RS-485 Control, 1 Ph	A to C
BAYEVAC10LG1	Electric Heater, 10kW, Lugs, RS-485 Control, 1 Ph	A to C
BAYEVC15BK1	Electric Heater, 15kW, Breaker, RS-485 Control, 1 Ph	B to C
BAYEVC20BK1	Electric Heater, 20kW, Breaker, RS-485 Control, 1 Ph	C
BAYEVCC25BK1	Electric Heater, 25kW, Breaker, RS-485 Control, 1 Ph	C
BAYEVAC10LG3	Electric Heater, 10kW, Lugs, RS-485 Control, 3 Ph	A to C
BAYEVC15LG3	Electric Heater, 15kW, Lugs, RS-485 Control, 3 Ph	B to C
BAYSUPFLGAA	Supply Duct Flange A	A
BAYSUPFLGBA	Supply Duct Flange B	B
BAYSUPFLGCA	Supply Duct Flange C	C
BAYRETLGAA	Return Duct Flange A	A
BAYRETLGBA	Return Duct Flange B	B
BAYRETLGCA	Return Duct Flange C	C
BAYSRKIT100A	Side Return Kit	A to C
BAYFLR1620A	High Velocity Filter Kit, 16" x 20" x 1" (10 filters)	A
BAYFLR2020A	High Velocity Filter Kit, 20" x 20" x 1" (10 filters)	B
BAYFLR2220A	High Velocity Filter Kit, 22" x 20" x 1" (10 filters)	C
TASB175SB	Plenum Stand with integrated sound baffle A	A
TASB215SB	Plenum Stand with integrated sound baffle B	B
TASB235SB	Plenum Stand with integrated sound baffle C	C
MITISRKIT01A	Side Return Kit with 16" x 20" Filter	A to C
BAYFRKIT175	Front Return Kit for 17.5" Cabinet	A
BAYFRKIT210	Front Return Kit for 21.0" Cabinet	B
BAYFRKIT235	Front Return Kit for 23.5" Cabinet	C
BAYBAFKT175A	Sound Baffle Kit for 17.5" Cabinet	A
BAYBAFKT215A	Sound Baffle Kit for 21.0" Cabinet	B
BAYBAFKT235A	Sound Baffle Kit for 23.5" Cabinet	C
TASSBK175	Sound Baffle Kit for 17.5" Cabinet	A
TASSBK210	Sound Baffle Kit for 21.0" Cabinet	B
TASSBK235	Sound Baffle Kit for 23.5" Cabinet	C
BAYICSKIT01A	Internal Condensate Switch Kit	A to C
BAYHHKIT001A	Horizontal Hanger Kit	A to C
BAYUVCLK001A	UV Lights	A to C
BAYLVKIT100A	Low Voltage Conduit Entry Kit	A to C
BAYSPEKT200A	Single Power Entry Kit	A to C
BAYWVAA05SC1AA	Hydronic Coil — 50,000 BTUH — Slide-in with control	A to A
BAYWVBB07SC1AA	Hydronic Coil — 70,000 BTUH — Slide-in with control	B to B
BAYWVCC08SC1AA	Hydronic Coil — 80,000 BTUH — Slide-in with control	C to C
BAYWACC11SC1AA	Hydronic Coil — 100,000 BTUH — Add on	C to C
BAYWVBRD485A	RS-485 Control for BAYWACC11SC1AA	C to C
BAYINSKT175A	Solcoustic® Liner Kit for 17.5" cabinet	A
BAYINSKT215A	Solcoustic® Liner Kit for 21.5" cabinet	B
BAYINSKT235A	Solcoustic® Liner Kit for 23.5" cabinet	C
BAYCNDPIP01A	3/4" PVC Threaded Pipe Kit foam Seal (10 per box)	A to C

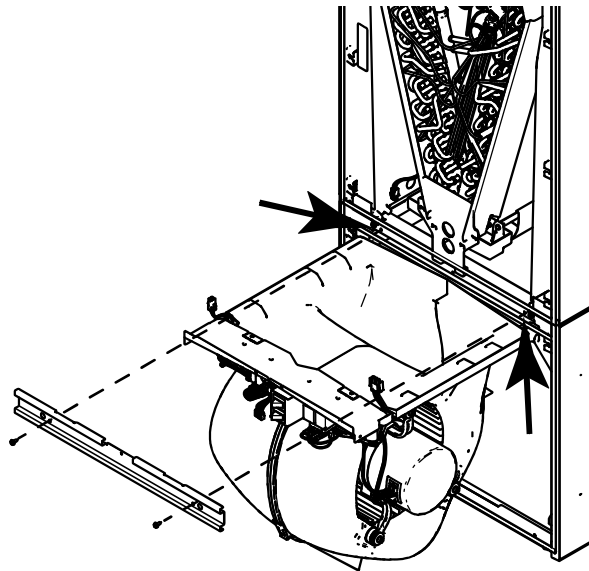
Optional Cabinet Disassembly

Note: If the unit must be transported in a horizontal position, it must be laid on its back (marked "REAR" on carton).

Note: To reassemble cabinet, follow the steps in reverse order. Ensure electrical connections are secure and the plug clips are engaged.

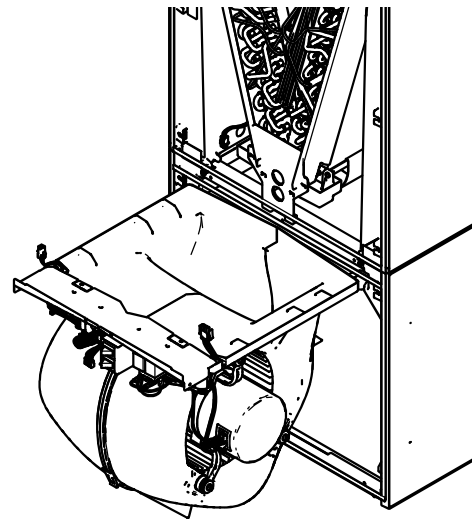
1. Remove all four front panels.
2. Remove the two screws on the seal bar and pull the seal bar straight out.
3. Disconnect all wiring connections routed to the blower assembly.

See [Table 2, p. 5](#)



4. Slide Blower assembly out of unit using built-in blower support channels and set aside.

Note: Remove the cardboard from the bottom of the blower. Cut the tie wrap and remove the foam block located at the motor.

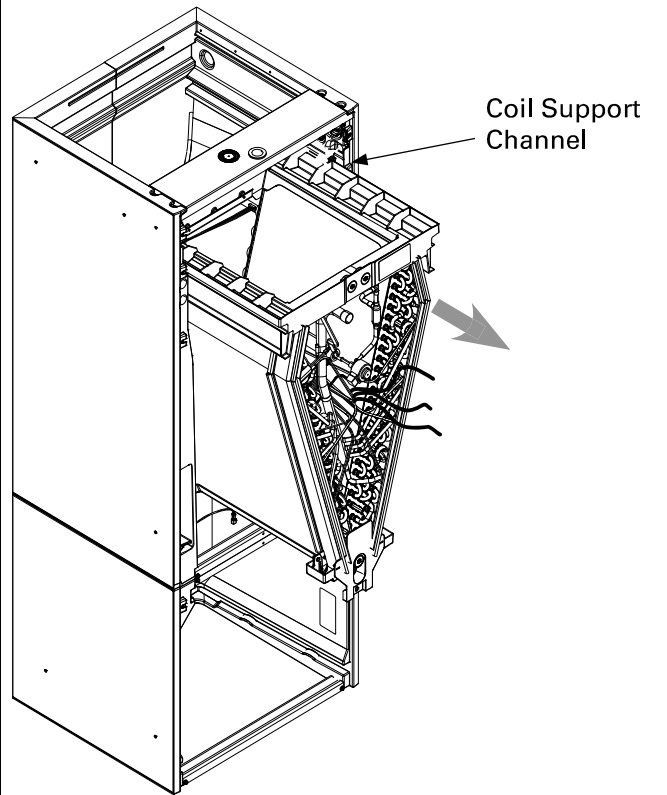


5. Disconnect wires to the EEV motor and sensors. Cut the wire ties on those wire harnesses if necessary and replace after re-installing.

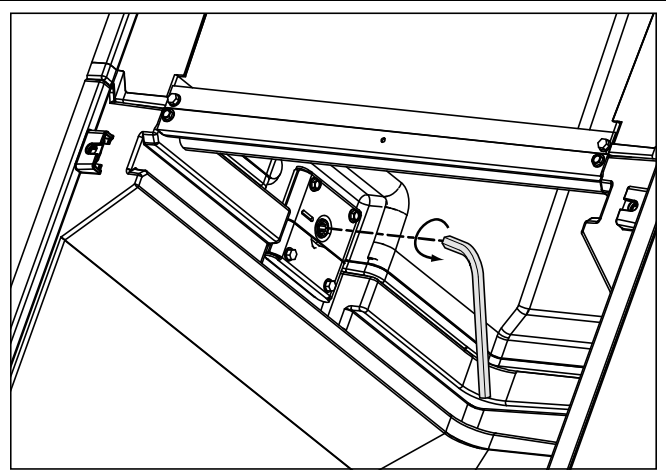
Note: If cut, wire ties that held the sensor must be replaced after the coil is placed back into the cabinet.

Optional Cabinet Disassembly

6. Slide Coil assembly out of unit using built-in coil support channels and set aside

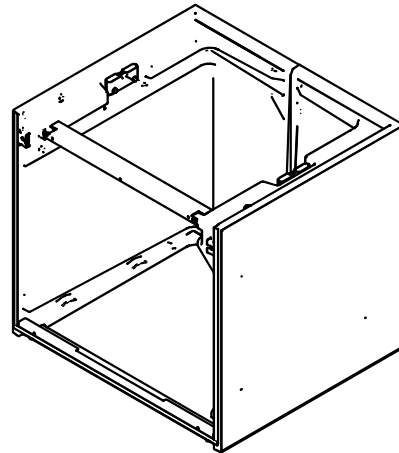
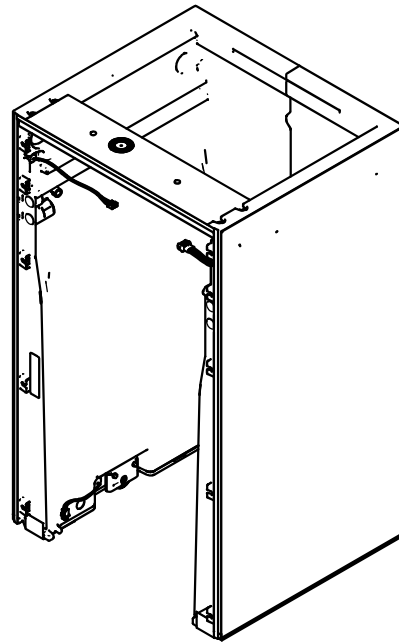


7. Use a 5/16 Allen wrench on the locking mechanism on each side of the bottom half of the cabinet to loosen the locking mechanism. The locks loosen by turning counter-clockwise approximately 3/4 of a turn.



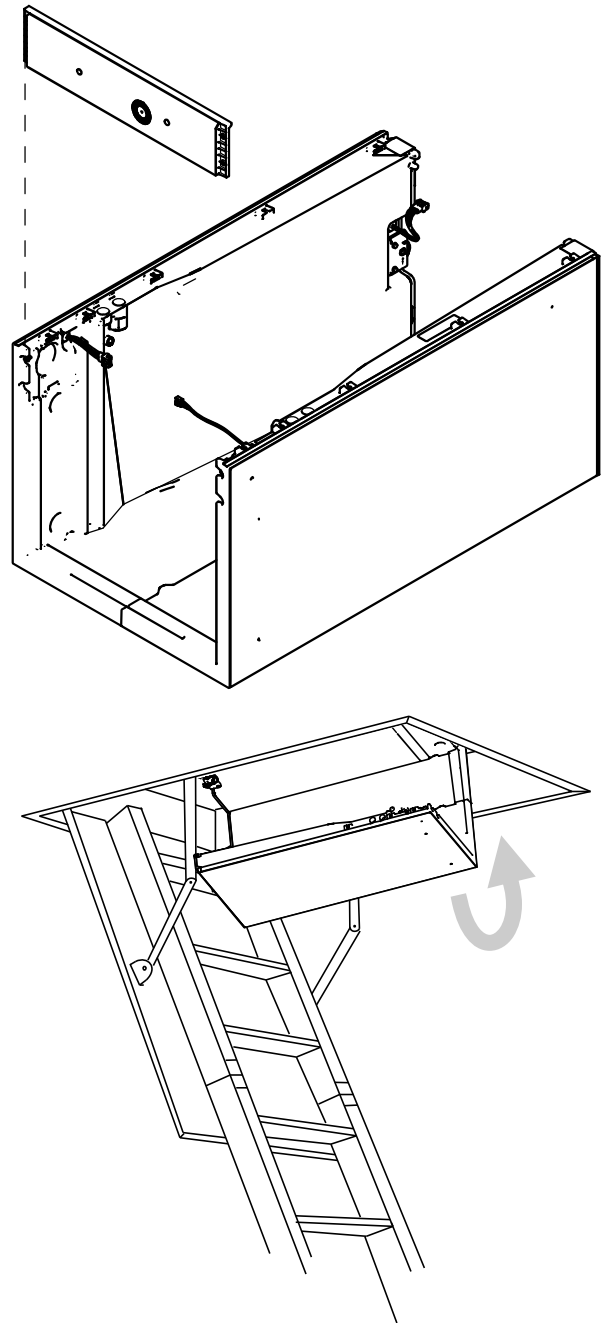
8. Lift the Coil section up and away from the Blower section. Set aside.

Note: When separating the two cabinet pieces, make sure the gasket remains intact.



Optional Cabinet Disassembly

9. For extremely tight spaces where the cabinet needs to be rotated through a small opening, remove the top panel and all cross members. Use a manual driver to avoid stripping screw holes.

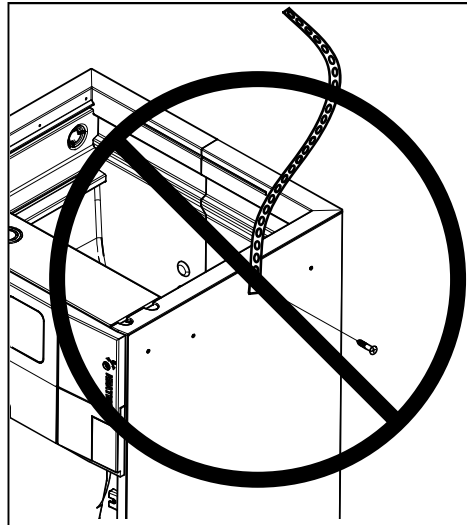


10. Continue preparation by following the proper carrying procedures shown in the next section.

Placing Unit at Location

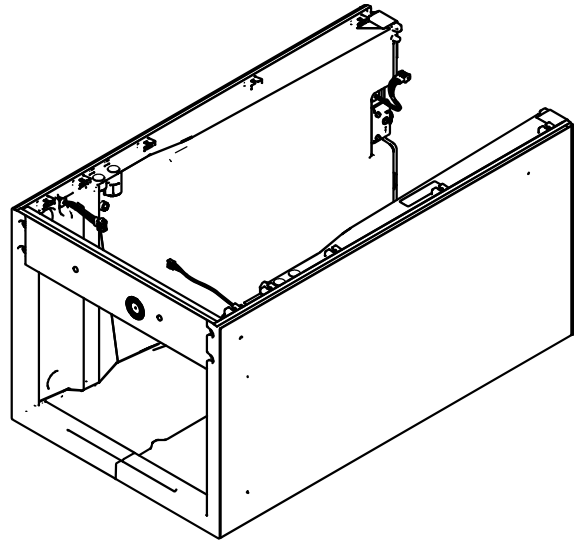
1. Carry the unit to the installation location
2. Reassembly by reversing the steps listed in Section 4 if disassembly was required. If cut, wire ties that held the sensor wiring must be replaced.

Important: Under no conditions should metal strapping be attached to the unit to be used as support mechanisms for carrying or suspension purposes.

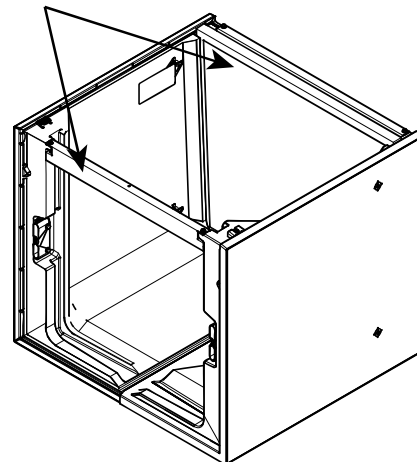


Approved Carrying:

- Hold by the cross members within the cabinet or unit top plate and use as handles for lifting and carrying the coil and blower sections.



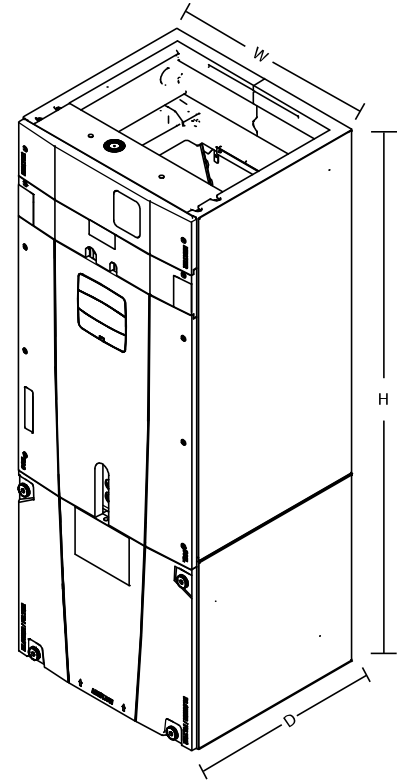
Cross Members



Unit Location Considerations

Table 3. Unit Dimensions and Weight

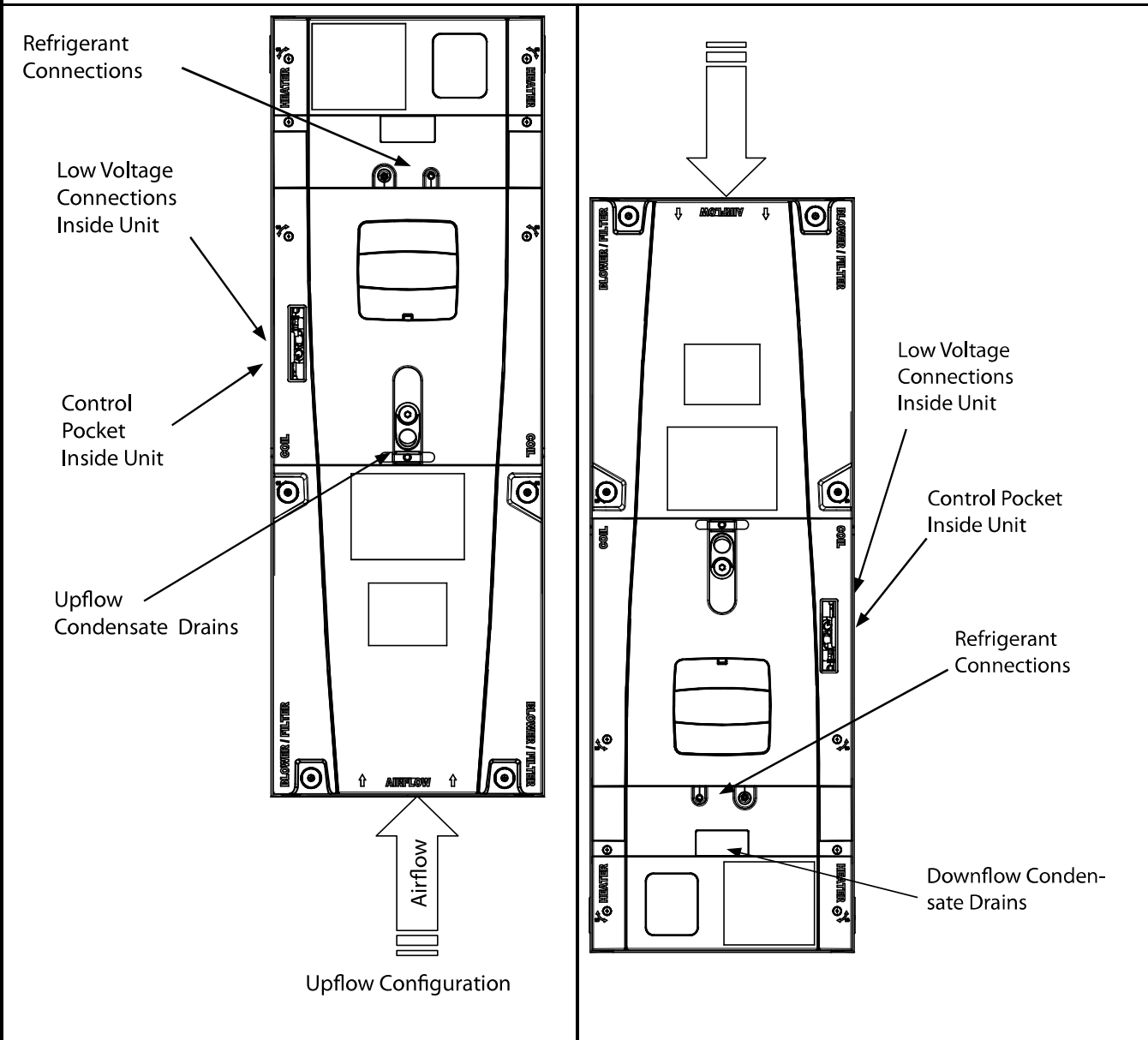
MODEL NUMBER	H x W x D (inches)	Coil and Heater Compartment Height * (inches)	Unit Net Weight (pounds)
TAM7A0A24H21SD TAM7A0A24H21EA	49.9 x 17.5 x 21.8	28.1	120
TAM7A0B30H21SD TAM7A0B30H21EA	55.7 x 21.3 x 21.8	33.9	133
TAM7A0C36H31SD TAM7A0C36H31EA	56.9 x 23.5 x 21.8	35.1	143
TAM7A0C42H31SD TAM7A0C42H31EA	56.9 x 23.5 x 21.8	35.1	158
TAM7A0C48H41SD TAM7A0C48H41EA	61.7 x 23.5 x 21.8	39.9	174
TAM7B0C60H51SC TAM7B0C60H51EA	61.7 x 23.5 x 21.8	39.9	178
* Blower compartment height is 21.8 inches.			



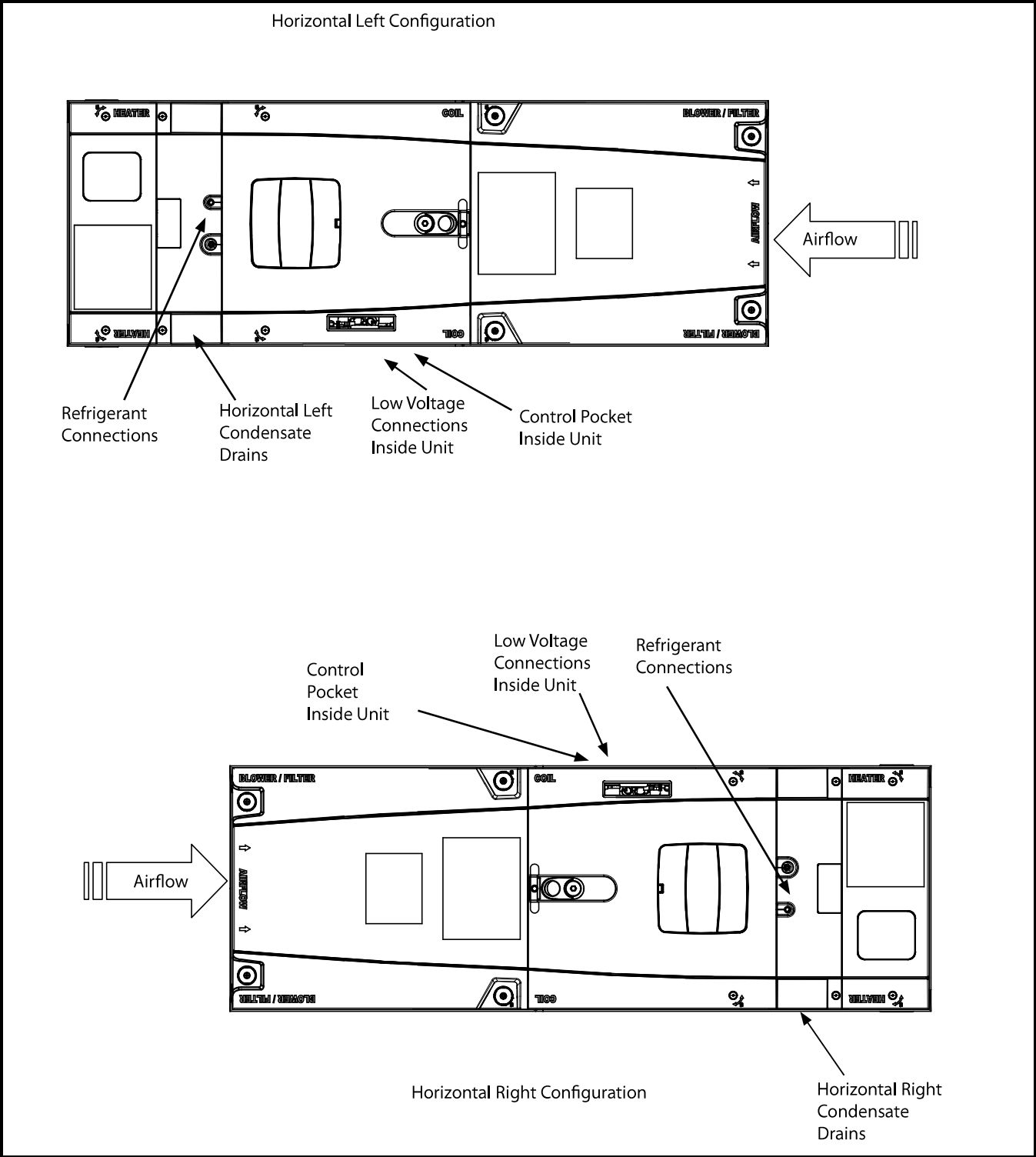
Four-Way Conversion

To place the unit in the configuration your application requires (upflow, downflow, horizontal right, or horizontal left), simply turn the unit to that orientation. Remember to adjust the badge accordingly.

Note: The air handlers are shipped from the factory suitable for four-way application.



Four-Way Conversion



Ducted and Non-Ducted Applications

Table 4. Non-Ducted Applications

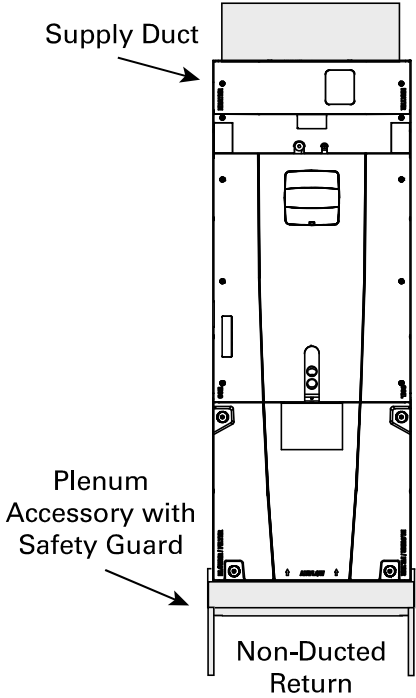
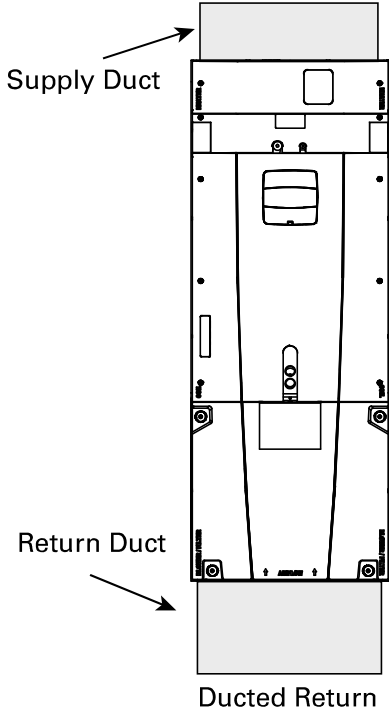
⚠ CAUTION HAZARDOUS VAPORS! Hazardous vapors can be distributed throughout the conditioned space and equipment damage can result. Do not install an air handler with a non-ducted return in the same closet, alcove, or utility room as a fossil fuel device.	 Supply Duct Plenum Accessory with Safety Guard Non-Ducted Return
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Table 5. Ducted Return Installations.

Ducted Return Installations: Installation in an attic, garage, or crawl space with ducted supply and return air is appropriate. Minimum clearances to combustible materials and service access must be observed (see outline drawing).	 Supply Duct Return Duct Ducted Return
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Additional Unit Preparation Considerations

For proper installation the following items must be considered prior to moving the unit to its installation site:

- Pursuant to Florida Building Code 13-610.2A.2.1, this unit meets the criteria for a factory sealed air handler.
- If a side return is needed for your application, the side return MUST be prepared prior to moving the air handler to its installation location. See the Side Return Kit #BAYSARKIT100A Installer Guide for detailed instructions, if used.
- When the air handler is located adjacent to the living area, the system should be carefully designed with returns which minimize noise transmission through the return air grill. Although the air handler is designed with large blowers operating at moderate speeds, any blower moving a high volume of air will produce audible noise which could be objectionable when the unit is located very close to a living area. It is often advisable to route the return ducts under the floor through the attic. Such design permits the installation of air return remote from the living area (i.e. central hall).
- Study the unit's outline drawing and dimensions prior to selecting the installation site. Note in advance which electrical conduit entry points and condensate drain holes are to be used, so that proper clearance allowances can be made for installation and future maintenance.
- Installation of the air handler must be made prior to , or at the same time as, the installation of the outdoor unit in order to allow access for refrigerant lines.

- Consider the overall space needed when external accessories are used, additional height and width requirements may exist.
- These units are not approved for outdoor installation.
- These units must be installed in the proper air flow direction.
- Any third-party heater accessories or hydronic coils must be downstream of the unit.

Note: No atomizing style humidifier is allowed in the return plenum with the use of this unit.

- Excessive bypass air may cause water blow-off, which will adversely affect system operation and air cleaner performance. To verify bypass airflow, follow the Bypass Humidifier Pre-Installation Checkout and Set-Up Procedures available through your local distributor. Ask for publication number 18-CH37D1-* Steam and Flow-through Fan Power Duct-mounted Humidifiers. Follow the humidifier installation instructions. These should only be installed on the supply air side of the system.

Note: The air handlers have been evaluated in accordance with the Code of Federal Regulations, Chapter XX, Part 3280 or the equivalent. "SUITABLE FOR MOBILE HOME USE."

Note: This unit is certified to UL 1995. The interior cabinet wall meets the following:

- UL94-5VA Flame Class Listed
- UL723 Steiner Tunnel Listed for 25/50 Flame/Smoke
- UL746C Listed for Exposure to Ultraviolet Light, Water Exposure and Immersion

Setting the Unit — Vertical Installation

Table 6. Considerations

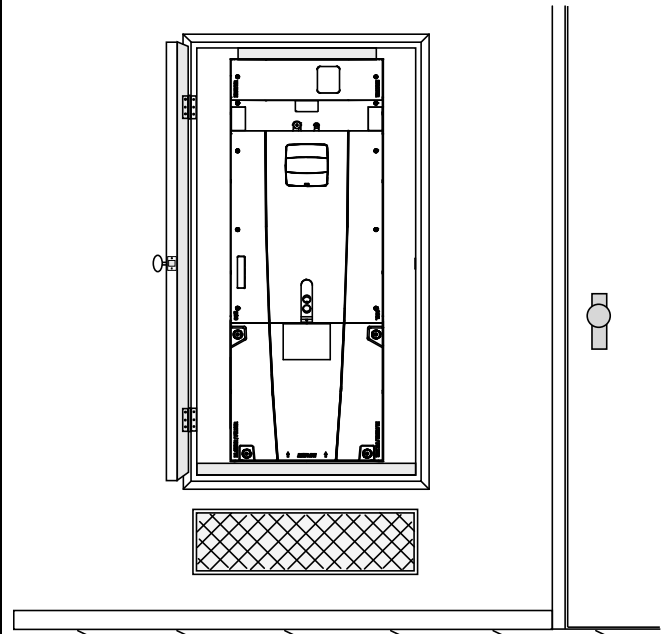
Provide a minimum height of 14 inches for proper unrestricted airflow below the unit. Allow a minimum of 21 inches clearance in front of the air handler to permit maintenance and removal of filter. • Position unit on suitable foundation. If a manufacturer approved accessory is not used, a frame strong enough to support the total weight of the unit, accessories, and duct work must be provided. • Isolate unit from the foundation using a suitable isolating material. Note: The following sound insulation kits are available to lessen objectionable sound. BAYINSKT175A for use with 17.5" cabinets BAYINSKT215A for use with 21.5" cabinets BAYINSKT235A for use with 23.5" cabinets	 The diagram shows a vertical unit installed in a closet. The unit is a tall, rectangular cabinet with a door on the left side. Inside the cabinet, there are various components including a fan, a filter, and a control panel. Below the unit, there is a rectangular area with a cross-hatch pattern, representing a sound insulation kit. The unit is mounted on a base. The closet door is on the right side, and the floor is indicated by a line at the bottom. The text 'Typical Closet Installation' is written below the diagram. Typical Closet Installation
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Table 7. Upflow Installation

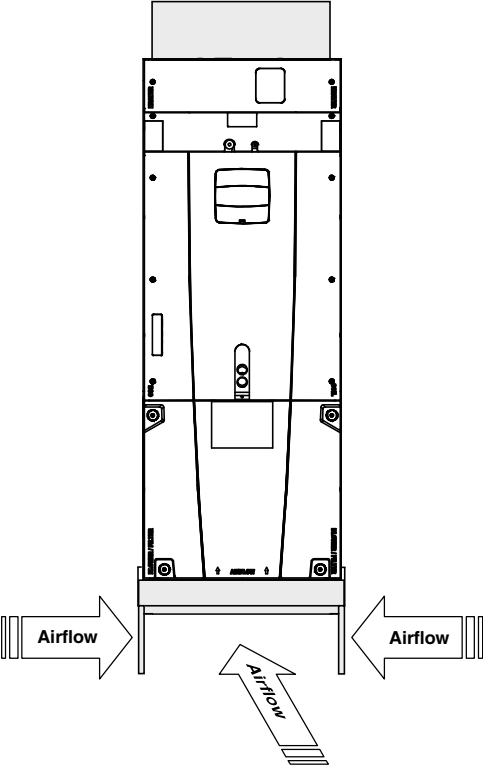
TASB Installation 1. Install the TASB plenum stand with integrated sound baffle using the TASB instructions. Note: Kit is used for open air applications. TASB175SB for use with 17.5" cabinets TASB215SB for use with 21.5" cabinets TASB235SB for use with 23.5" cabinets MITISRKIT1620 — Side return kit with 16" x 20" filter Contact your distributor for more information. Note: The following sound insulation kits are available to lessen objectionable sound. BAYINSKT175A for use with 17.5" cabinets BAYINSKT215A for use with 21.5" cabinets BAYINSKT235A for use with 23.5" cabinets	 The diagram shows a vertical unit with a TASB plenum stand installed at the bottom. Three arrows labeled 'Airflow' point upwards from the bottom into the unit, indicating the direction of air flow. Typical TASB Installation
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Table 8. Plenum Installation

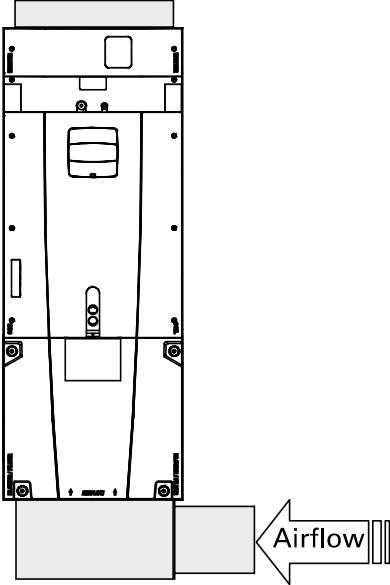
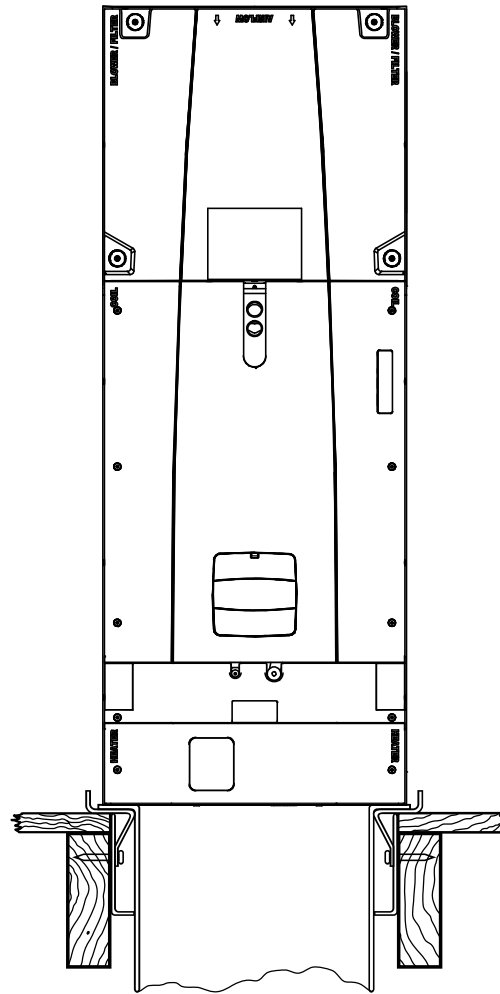
2. Assemble the plenum using the plenum's Installer Guide. On units with sheet metal returns: Return plenum must be flanged. Sheet metal drill point screws must be 1/2" in length or shorter.	 The diagram shows a vertical unit with a plenum installed at the bottom. An arrow labeled 'Airflow' points upwards from the plenum into the unit, indicating the direction of air flow. Typical Plenum Installation
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Table 9. Downflow Installation

- Downflow installation must comply with national, state, and local codes.
 - The air handlers are rated for zero clearance from combustible materials.
3. Prepare the location site as appropriate for your application and per national, state, and local code requirements.
 4. Set the unit in position.



Representative Illustration
Typical Downflow Installation

Setting the Unit — Horizontal Installations

Table 10. Considerations

Important: Due to the unique design of this unit, which allows the electrical wiring to be routed within the insulation, do not screw, cut, or otherwise puncture the unit cabinet in any location other than the ones illustrated in this Installer Guide or in an approved accessory's Installer Guide.

Important: Make certain that the unit has been installed in a level position to ensure proper draining.

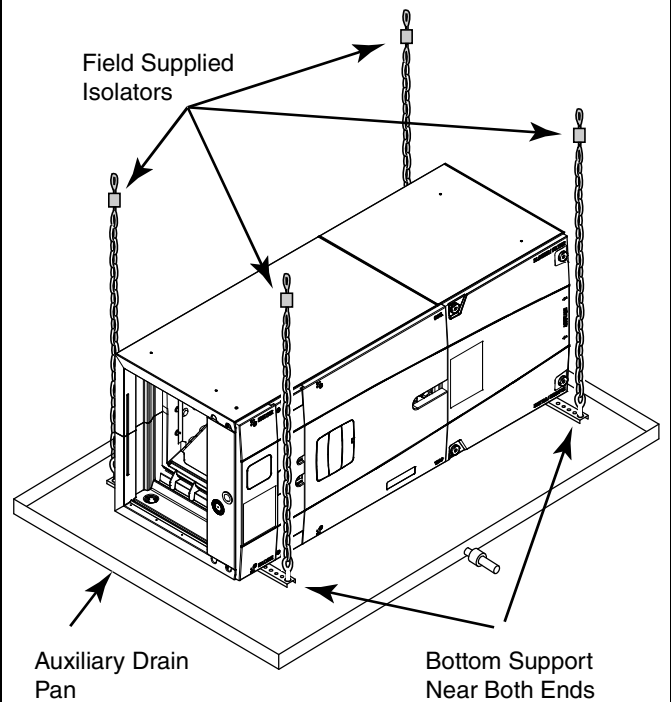
Important: Under no conditions should metal strapping be attached to the unit to be used as support mechanisms for carrying or suspension purposes.

1. Support the unit from the bottom (near both ends). The service access must remain unobstructed.
 - a. Approved bottom support methods are rail, u-channels (Unistrut®), or other load bearing materials.
 - b. The unit must be isolated carefully to prevent sound transmission. Field supplied vibration isolators are recommended.

Important: The unit can only be supported from the bottom unless using kit BAYHHKIT001A. Do not drill or screw supports into any area of the cabinet.

Note: Do not allow the unit to be used as strain relief.

2. Install an auxiliary drain pan under the horizontal air handler to prevent possible damage to ceilings.
3. Install an auxiliary drain pan under the horizontal air handler to prevent possible damage to ceilings.
 - a. Isolate the auxiliary drain pan from the unit and from the structure.
 - b. Connect the auxiliary drain pan to a separate drain line and terminate according to local codes.



Note: BAYHHKIT001A Hanging Bracket Kit may be ordered separately.

Important: The BAYHHKIT001A may not be used if the cabinet has been altered per Installer Guide 18-GJ58D1-1

Connecting the Duct work

Table 11. Duct Connection Considerations

Important: Due to the unique design of this unit, which allows the electrical wiring to be routed within the insulation, do not screw, cut, or otherwise puncture the unit cabinet in any location other than the ones illustrated in this Installer Guide or in an approved accessory's Installer Guide.

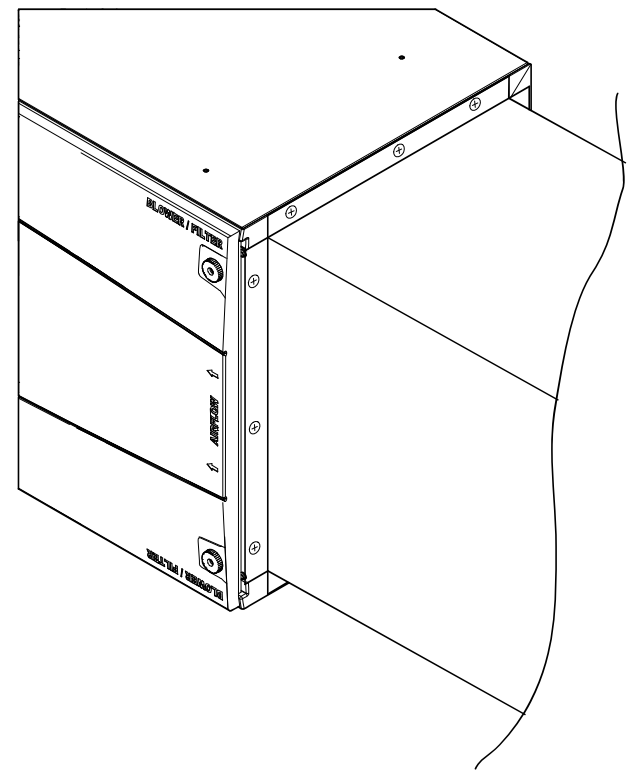
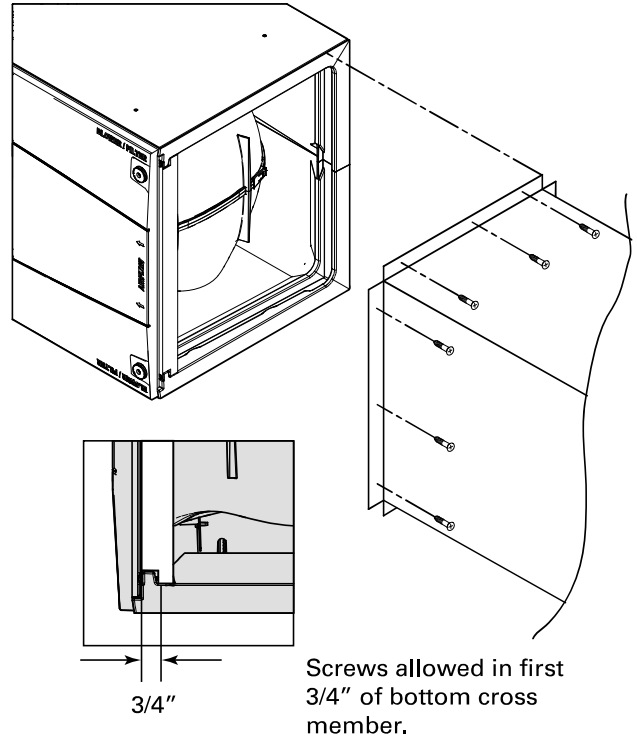
Important: Under no conditions should metal strapping be attached to the unit to be used as support mechanisms for carrying or suspension purposes..

Important: On units with sheet metal returns: Return air plenum must be flanged. Sheet metal drill point screws must be 1/2" in length or shorter.

- The supply and return air ducts must be connected to the unit with non flammable duct connectors.
- See the Outline drawing for sizes of the duct connections.
- After the ducts are secured, seal around the supply and return ducts to prevent air leakage.
- Insulate all duct work that will be outside of conditioned spaces.
- Convertible Duct Flange Kits are available to connect the supply plenum or for mounting on the discharge opening to provide a "flush fit" for 1–1/2" duct board applications.
- If front or rear return is required, the air handler must be elevated — placed on a pedestal or plenum and duct must be connected to this pedestal or plenum.
- If side return is required, the Side Return Kit # BAYSRKIT100A accessory must be used. A remote filter will be required.
- To ensure maximum efficiency and system performance, the existing supply and return duct system static pressures must not exceed the total available static pressure of the air handler. Reference ACCA Manual D, Manual S and Manual RS along with the air handler Product Data and Service Facts for additional information.

Note: Side return is not approved without Side Return Kit # BAYSRKIT100. More than one Side Return Kit may be necessary depending on the application. Refer to the Installation Guide in BAYSRKIT100 for approved duct connections, sizing, number, transitions, and accessory application.

Note: Duct work must be supported as appropriate. See National and local codes for guidelines. Do not depend on the unit to support duct work.



Refrigerant Line

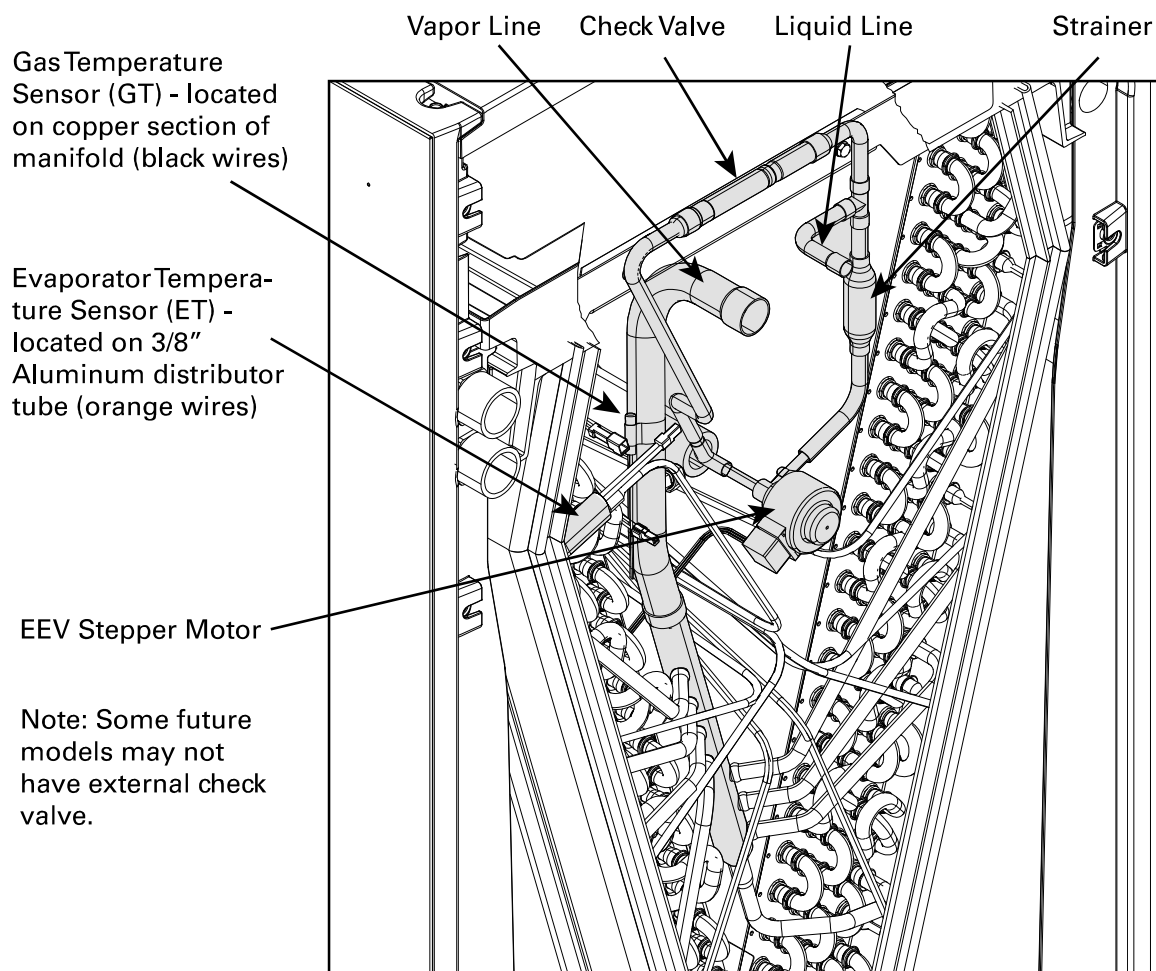
Table 12. Refrigerant Line Connection Sizes

Model	Vapor Line Connection	Liquid Line Connection
TAM7A0A24H21SD, TAM7A0A24H21EA	3/4	3/8
TAM7A0B30H21SD, TAM7A0B30H21EA	3/4	3/8
TAM7A0C36H31SD, TAM7A0C36H31EA	7/8	3/8
TAM7A0C42H31SD, TAM7A0C42H31EA	7/8	3/8
TAM7A0C48H41SD, TAM7A0C48H41EA	7/8	3/8
TAM7B0C60H51SC, TAM7B0C60H51EA	7/8	3/8

Notes:

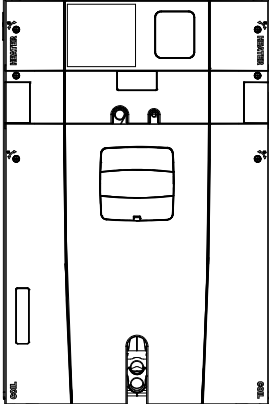
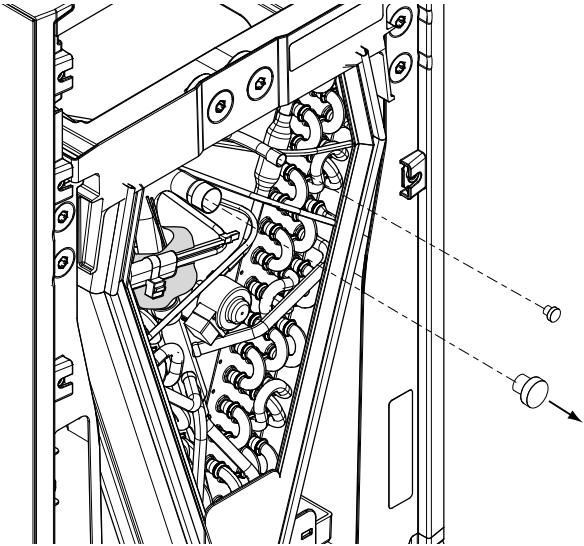
1. This table indicates the tubing connection diameters at the indoor coil. A field supplied reducing coupling may be required.
2. All AHRI listed systems are tested with 25 feet of refrigeration tubing; the rated tubing diameters are located in the electronic performance data system.
3. If the refrigeration lines exceed 60 feet in linear length and/or if alternate size refrigeration tubing is present at the job, please consult SS-APG006-EN or 32-3312** (latest version)

Refrigerant System Layout



Refrigerant Line Brazing

Table 13. Braze the Refrigerant Lines

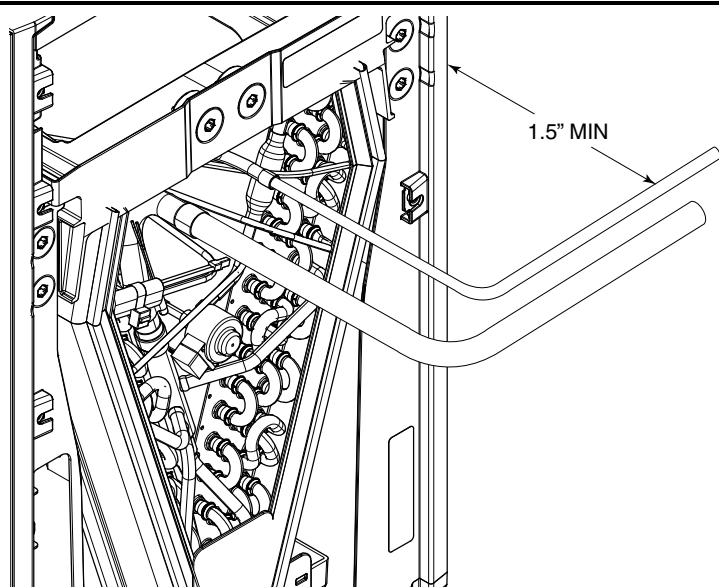
1. Remove Heater, Coil, and Line Set panels. See Table 2, p. 5	 Heater Panel 1 Line Set Panel 2 Coil Panel 3
Important: Do NOT unseal coil refrigerant connection stubs until ready to make connections. Important: Heat Sensitive Sensors. The Gas Temperature Sensor must be removed or a wet rag must be wrapped around the suction line between the Sensor and the braze joint to protect it from failure due to overheating. 2. Remove the sealing plug from the indoor coil suction line. 3. Remove the sealing plug from the indoor coil liquid line. <i>This coil is pressurized with 8–12 psig of dry air. Do not stand directly in front of the coil connections when removing sealing plugs.</i>	

Refrigerant Line Brazing

Table 13. Braze the Refrigerant Lines (continued)

4. Connect, but do not braze field line set to indoor coil.
 - a. Allow a minimum of 1.5 inches of refrigerant line set before using an elbow coupling.

Important: Service access to the auxiliary heater must remain unobstructed.

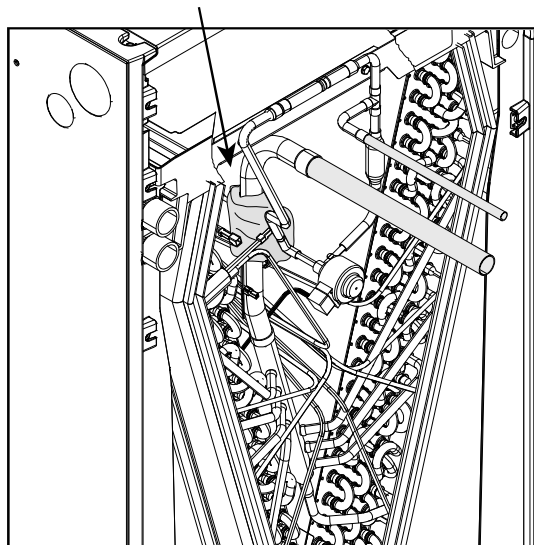


Important: Heat Sensitive Sensor. The temperature Sensor must be removed or a wet rag must be wrapped around the suction line between the Sensor and the braze joint to protect the Sensor from failure due to overheating.

5. Braze refrigerant line connections.
 - a. Pull back the insulation before brazing the suction line.
 - b. Wrap the Gas Temperature Sensor (GT) with a wet rag.
 - c. Braze the refrigerant line connections.

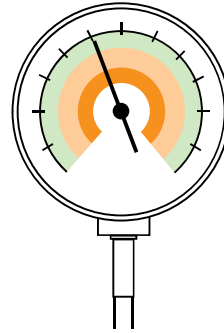
Note: The suction line must be insulated prior to brazing the line set to the air handler stubs.

Wet Rag on Gas Temperature Sensor (GT)

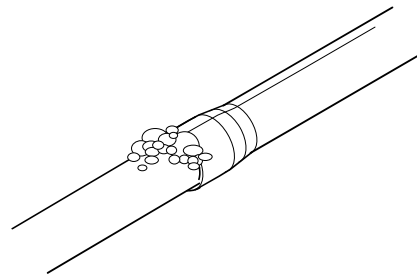


6. Pressurize the refrigerant lines and evaporator coil to 150 PSIG using dry nitrogen.

150 PSIG

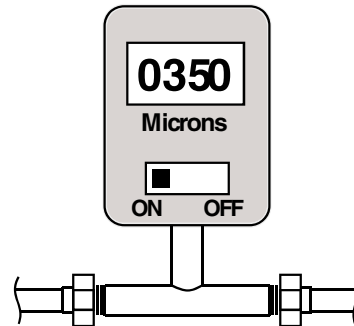


7. Check for leaks by using a soapy solution or bubbles at each brazed location.



Important: Do not open the service valves until the refrigerant lines and indoor coil leak check and evacuation are complete.

8. Evacuate until the micron gauge reads no higher than 350 microns, then close off the valve to the vacuum pump.



9. Observe the micron gauge. Evacuation is complete if the micron gauge does not rise above 500 microns in one (1) minute.
- Once evacuation is complete blank off the vacuum pump and micron gauge, and close the valves on the manifold gauge set.

Note: Charge system using Outdoor unit's Installer Guide or Service Facts.

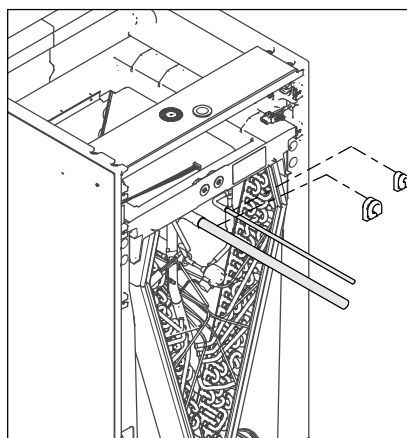
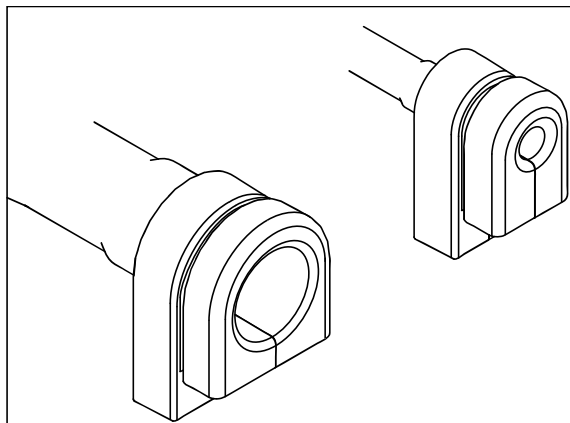
Note: Ensure to wipe any refrigerant oil off the panels.



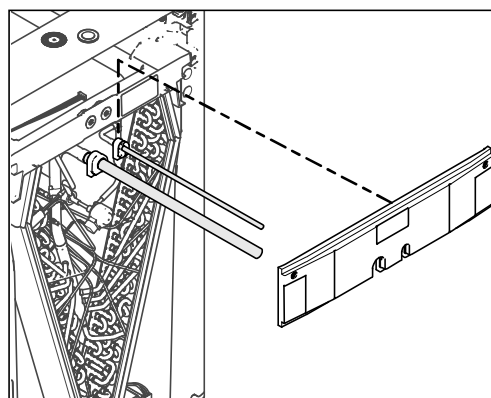
10. Replace the Line Set panel.

- a. Allow time for tubing to cool.
- b. Install grommets to line set piping in orientation shown.
- c. Allow time for tubing to cool.
- d. Install grommets to line set piping in orientation shown.

Note: A slight amount of dish soap can be used to aid in the installation of the grommets. Remove any excess from the tubing and grommet after the grommet is installed.



**SEE ENLARGED
ILLUSTRATION FOR ORIENTATION**



Condensate Drain Piping

Table 14. Condensate Drain Piping Considerations

• Condensate drain plumbing must comply with national, state, and local codes. • Route condensate drain lines away from air handler so they do not interfere with access panels. • Slope the drain lines downward a minimum of 1/4" per foot, support per local codes. • Do not use reducing fittings in the condensate drain lines.	• Do not connect the drain line to a closed drain system. • Do not use a torch or flame near the plastic drain pan coupling. • A P-trap is not required for proper drainage due to the positive pressure of the air handler; however, it is recommended to prevent efficiency loss of conditioned air.
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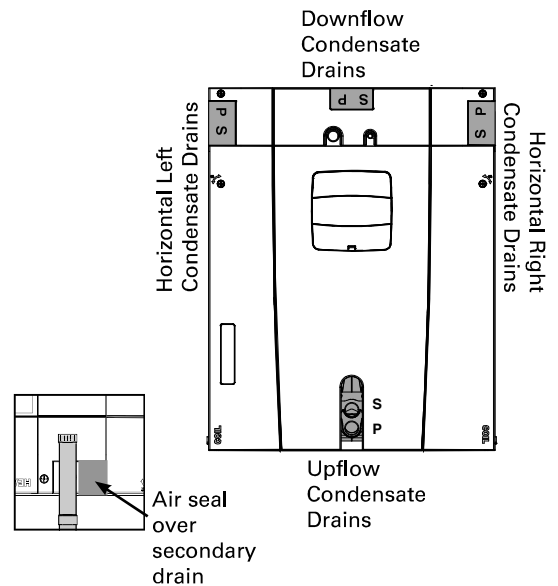
Condensate Drain Piping

Table 15. Connect Condensate Drain Piping

Note: Downflow and horizontal orientations require the Coil panel to be removed in order to make the drain connections.

Note: Make certain that the unit has been installed in a level position to allow for proper draining.

1. Select the drain connections that are oriented for your application.
2. Prepare the condensate drain connections.
 - a. From the factory, the unit comes with plugs in both upflow condensate drains and an additional plug in the documentation packet.
 - b. For upflow applications, remove upflow condensate plug(s) and connect condensate piping.
 - c. For all other applications, do not remove upflow condensate plugs. Remove the cover from the needed condensate drain connections and connect condensate piping.
 - d. If the secondary condensate opening is not used, plug the condensate opening with the fitting supplied in the documentation pack. Use scissors to cut the air seal in half and re-install over the unused opening.



Note: A small amount of sealant must be applied around the drain line (s) passing through the panel to prevent air leakage and possible water drips.

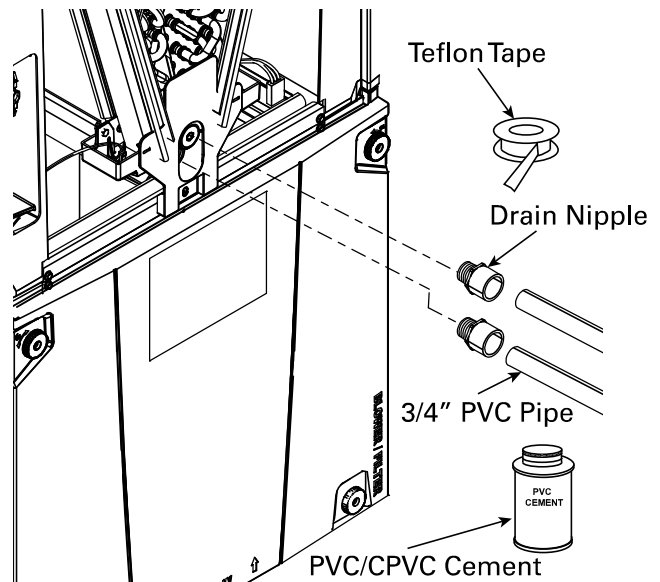
- Dry fit and test clearance for coil panel removal before applying PVC/CPVC cement.
 - Use Teflon tape on the air handler drain line connections. Do not use pipe joint compound or PVC/CPVC cement on drain nipple.
 - Hand tighten the drain pipe.
3. For upflow installations, connect 3/4" PVC pipe to the threaded drain nipple with PVC/CPVC cement. 3" minimum clearance to the condensate piping is needed for coil panel removal. Thread the assembly into the primary drain connection (repeat for the secondary drain connection if used).

IMPORTANT: For Horizontal and Downflow installations, the following order must be observed:

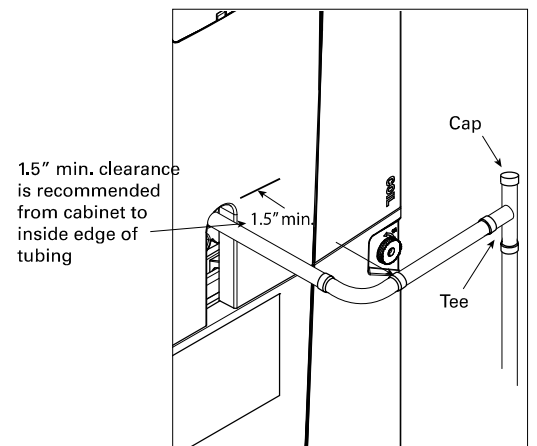
- a. Remove panel and insert the 3/4" nipples.
- b. Reinstall the panel.
- c. Connect the condensate lines to the nipples.

Important: For Horizontal and Downflow installations, the following order must be observed:

Note: A small amount of sealant must be applied around the drain line (s) passing through the panel to prevent air leakage and possible water drips.

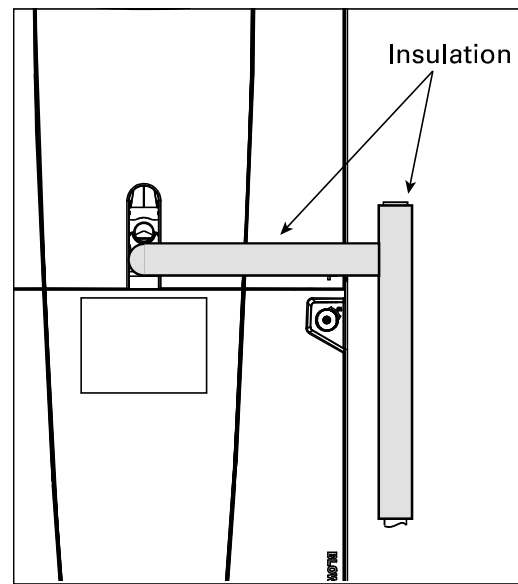


4. Install a clean-out tee in the primary drain line for future maintenance. It is recommended that you install a cap on the top of the tee.



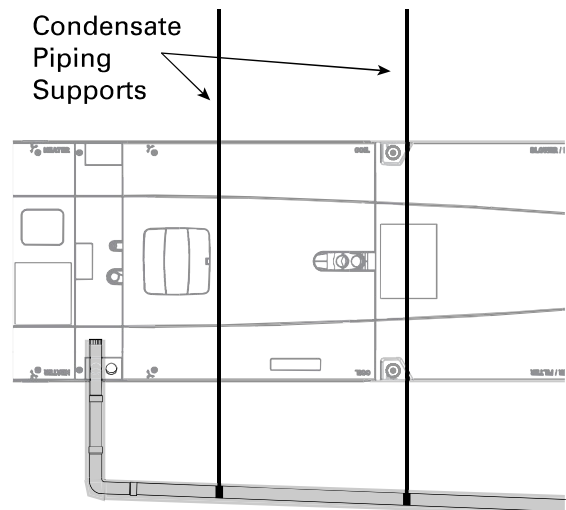
5. Insulate the primary drain line to prevent sweating where dew point temperatures may be met. (Optional depending on climate and application needs.)

Provide a means of drainage to prevent winter freeze-up of condensate line (Optional depending on climate and application needs).



6. Support the condensate piping outside the unit per local codes for proper drainage and to prevent sagging.

Allow 1/4" of downward slope for each foot of pipe.

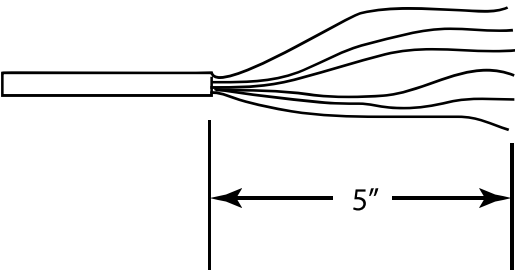
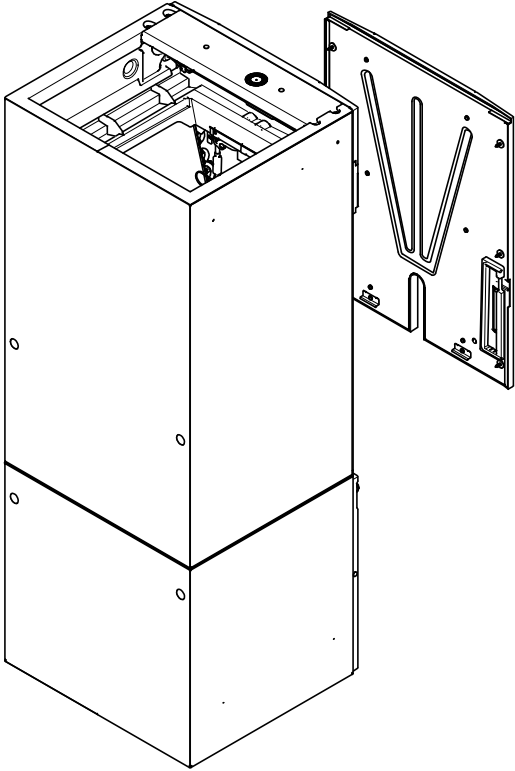


Electrical — Low Voltage

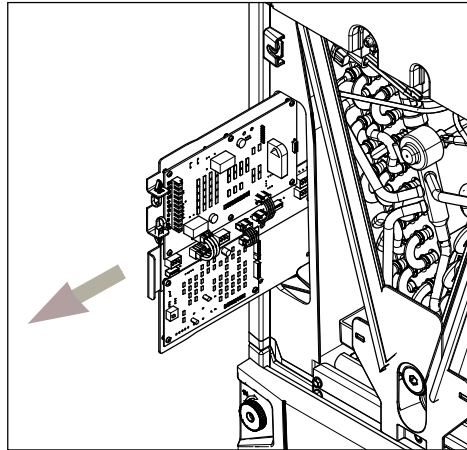
Table 16. Low Voltage Maximum Wire Length

The associated table defines the maximum total length of low voltage wiring from the outdoor unit to the indoor unit, and to the thermostat.	24 VOLTS	
	WIRE SIZE	MAX. WIRE LENGTH
	18 AWG	150 FT.
	16 AWG	225 FT.
	14 AWG	300 FT.

Table 17. Low Voltage Hook-up Instructions

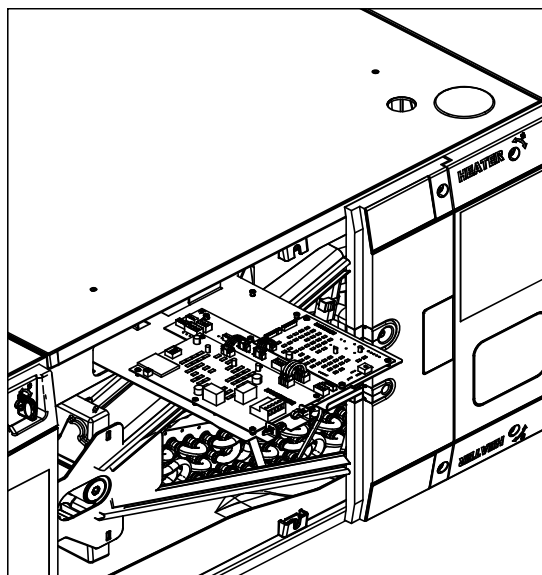
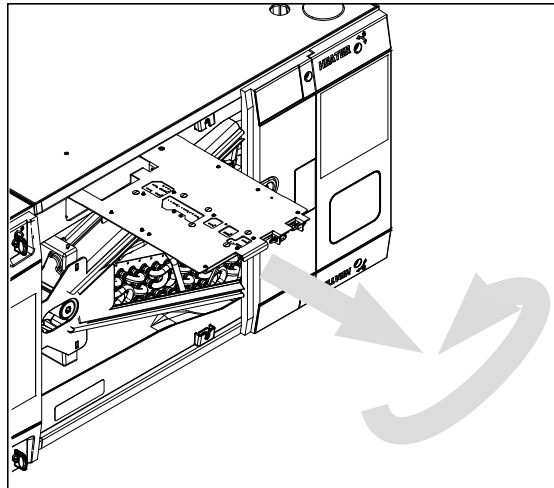
Note: Strain relief must be provided on the inside of the air handler cabinet for the low voltage wiring. Field supplied thermostat wired may be wire tied as a bundle to the existing strain relieved low voltage pigtail leads in the air handler unit. 1. Route control wiring to unit. Remove the external sheathing of the wiring approximately 5". Note: Optional Low Voltage Conduit Entry Kit number BAYLVKIT100A is available.	
2. Remove Coil panel by turning the Phillips head door fasteners, rotating the door away from the cabinet, and removing.	

3. Remove the control board from the control pocket by sliding the control pocket mounting plate out until the first stop is reached.

**For Horizontal Right Installations Only**

Remove the control board completely from the control pocket. Rotate the control board 180 degrees. Place the control board 2–3 inches back into the control pocket. Attach wiring per appropriate hook-up diagram or perform service.

Important: After wires have been connected or service performed, the control board **MUST** be rotated back to the original orientation before inserting into cabinet control pocket.

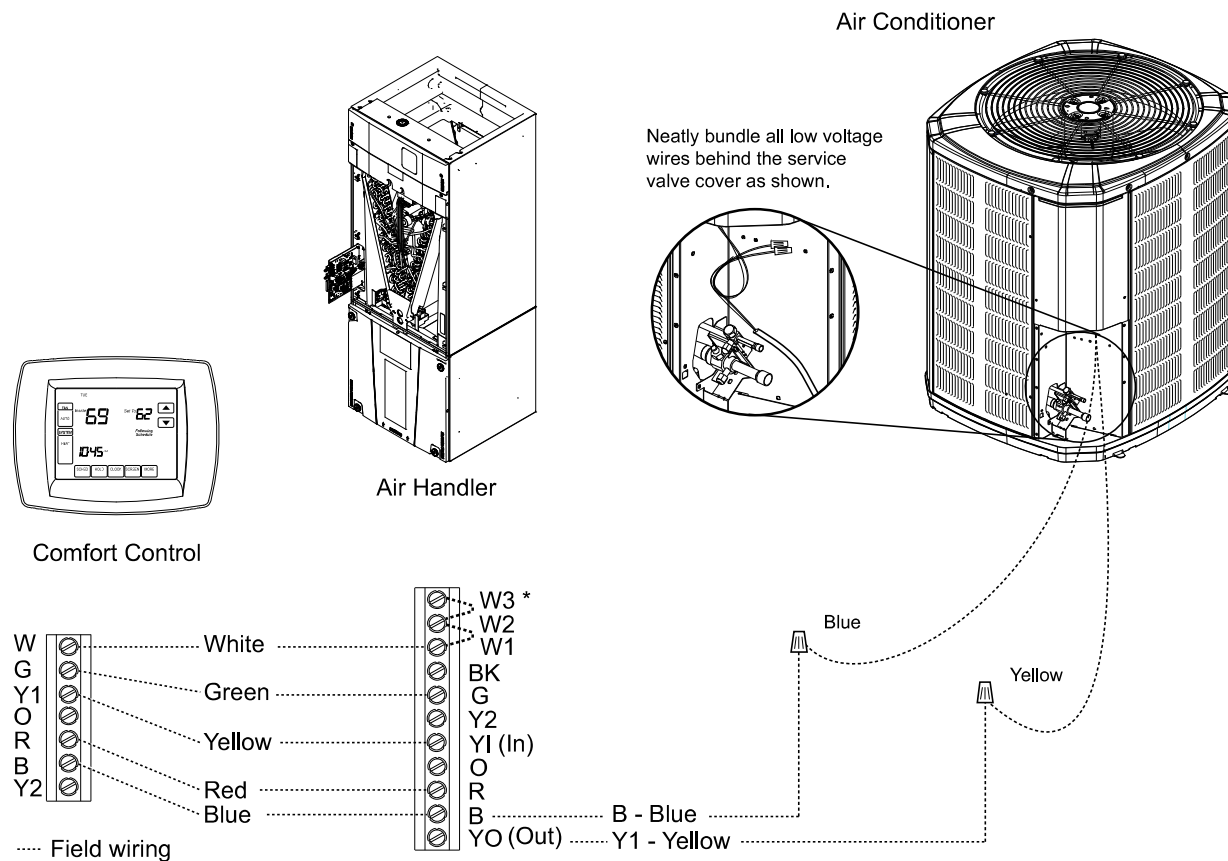


4. Make connections per hookup diagrams.



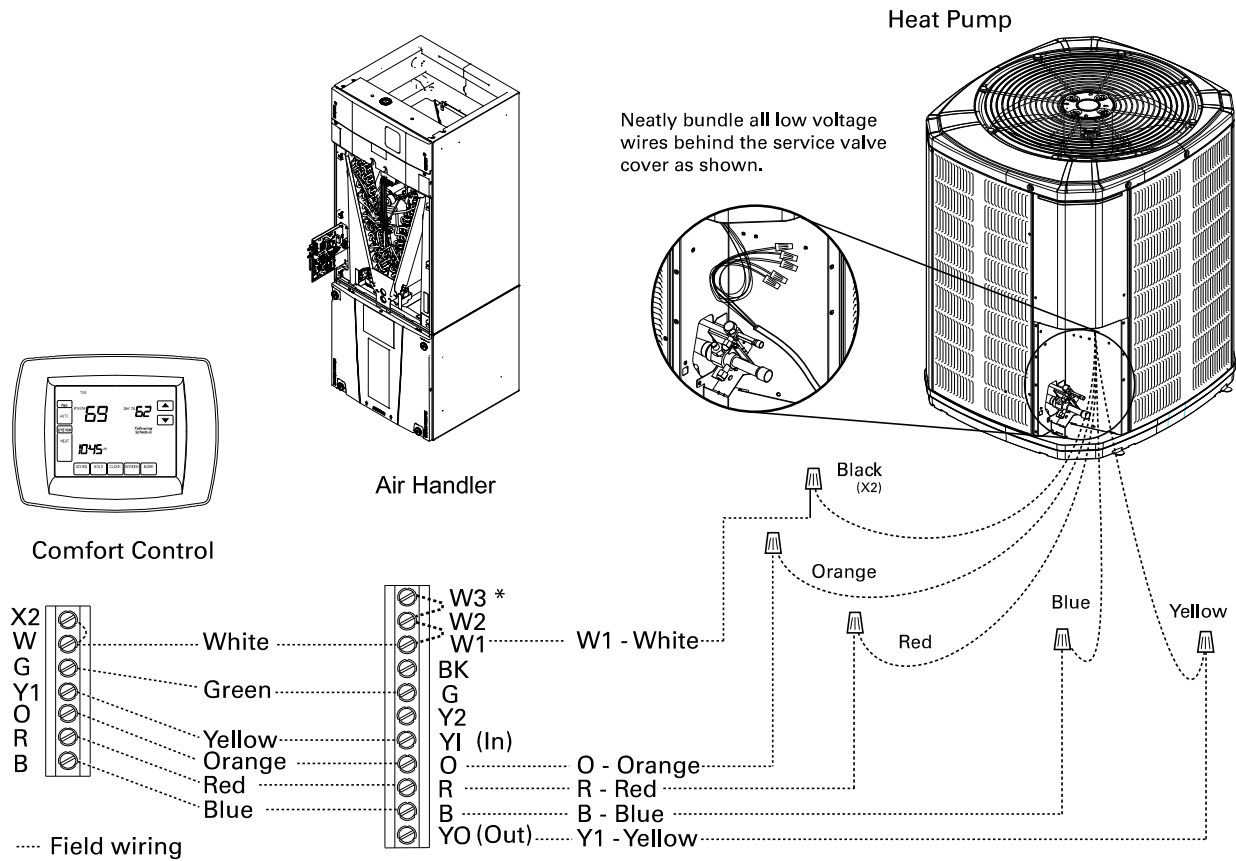
Scan to see a
video on Field
Wiring the TAM7

Figure 1. Field Wiring — Single Stage Cooling



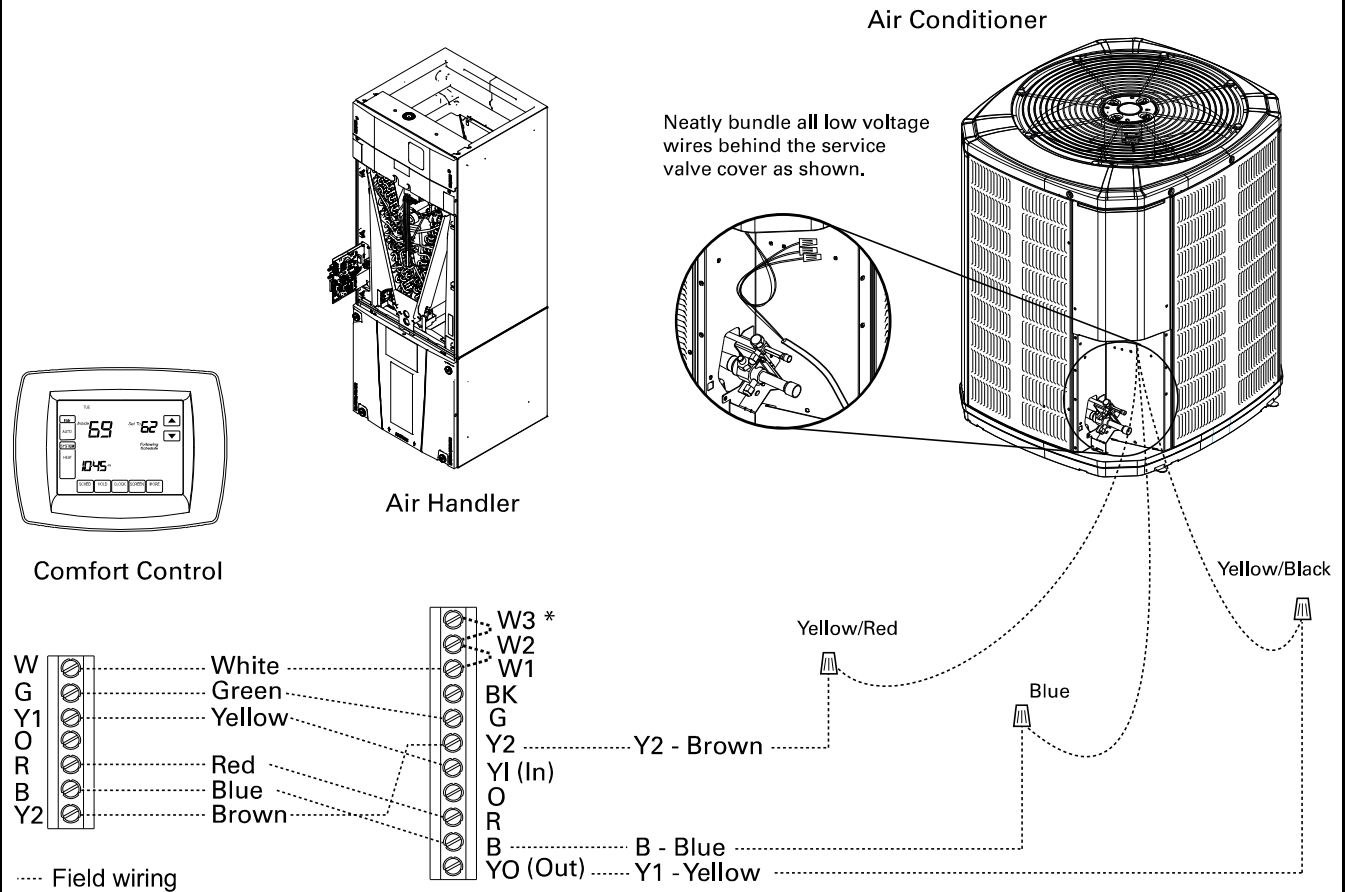
- * For multiple stages of electric heat, jumper W1, W2, and W3 together if control has only one stage of heat.
- Y1 and Y0 connections must be made as shown for freeze protection and internally mounted condensate overflow circuits to work properly. OD unit pump down may occur if these connections are not made correctly.
- Internally mounted condensate switch is optional and must be ordered separately.
- If 3rd party condensate overflow switches are installed, they should be wired between Y of the thermostat and Y1 of the Airflow control.

Figure 2. Field Wiring — Single Stage HP



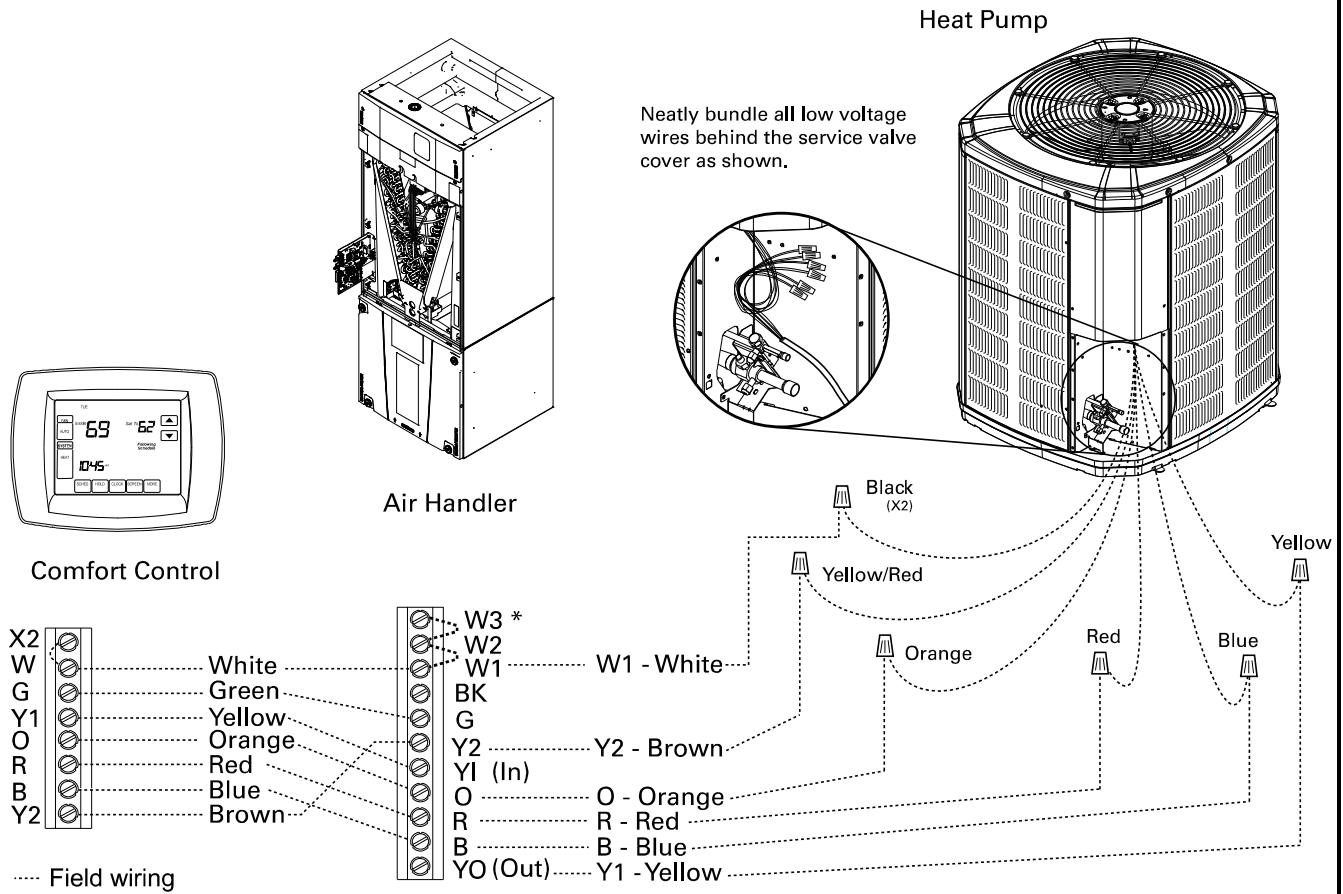
- * For multiple stages of electric heat, jumper W1, W2, and W3 together if control has only one stage of heat.
- Y1 and Y0 connections must be made as shown for freeze protection and internally mounted condensate overflow circuits to work properly. OD unit pump down may occur if these connections are not made correctly.
- Internally mounted condensate switch is optional and must be ordered separately.
- If 3rd party condensate overflow switches are installed, they should be wired between Y of the thermostat and Y1 of the Airflow control.

Figure 3. Field Wiring — Two Stage Cooling



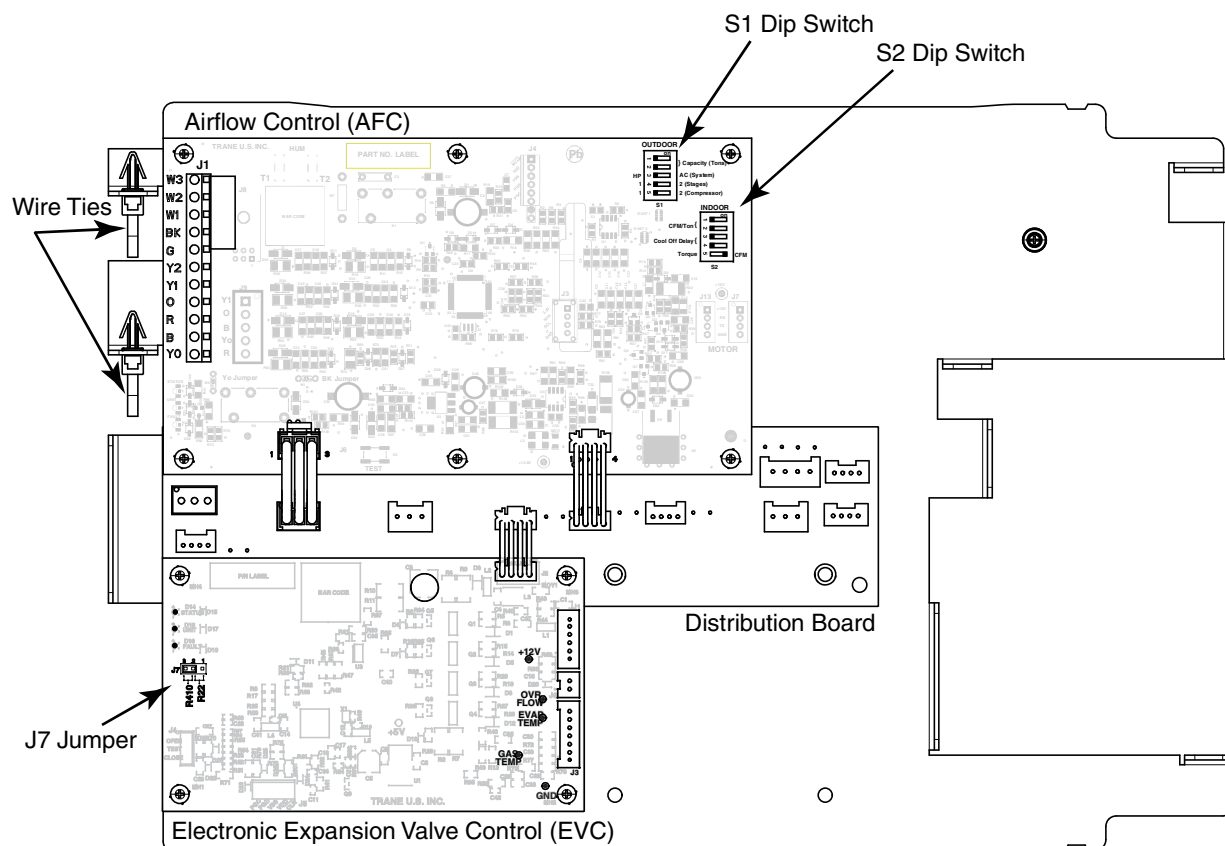
- * For multiple stages of electric heat, jumper W1, W2, and W3 together if control has only one stage of heat.
- Yi and Y0 connections must be made as shown for freeze protection and internally mounted condensate overflow circuits to work properly. OD unit pump down may occur if these connections are not made correctly.
- Internally mounted condensate switch is optional and must be ordered separately.
- If 3rd party condensate overflow switches are installed, they should be wired between Y of the thermostat and Y1 of the Airflow control.

Figure 4. Field Wiring — Two Stage HP



- * For multiple stages of electric heat, jumper W1, W2, and W3 together if control has only one stage of heat.
- Y1 and Y0 connections must be made as shown for freeze protection and internally mounted condensate overflow circuits to work properly. OD unit pump down may occur if these connections are not made correctly.
- Internally mounted condensate switch is optional and must be ordered separately.
- If 3rd party condensate overflow switches are installed, they should be wired between Y of the thermostat and Y1 of the Airflow control.

5. Secure the sheathed wiring to the control pocket mounting plate using the factory supplied wire ties attached to the tabs as shown.



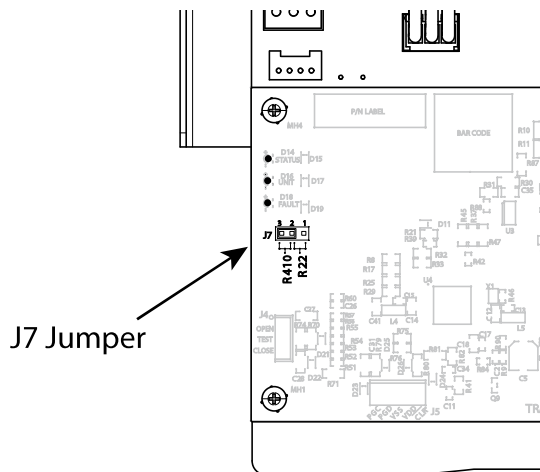
6. Set Jumper and Dip Switches

- Set the system refrigerant to either R410A or R-22 using the Refrigerant Jumper located on the Expansion Valve Control board (EVC) in the Control Pocket.
- Factory default is R410A.

Note: The power must be shut off and then re-applied in order for the EVC to recognize the change.



Scan to see a video on setting the TAM7 DIP Switches.



Electronic Expansion Valve Control (EVC)

Dip Switches



Scan to see a video on setting the TAM7 DIP Switches.

[Click or Scan to see a video on setting the TAM7 DIP Switches](#)

Dip Switch S1

- Set the S1-1 and S1-2 dip switches for the OD multiplier (tonnage) per the chart.

INDOOR MODEL	S1-1	S1-2	OD MULTIPLIER
TAM7A0A24H21SD TAM7A0A24H21EA	OFF	OFF	2
	OFF	ON	1.5
	ON	OFF	2.5
	ON	ON	3
TAM7A0B30H21SD TAM7A0B30H21EA	OFF	OFF	2
	OFF	ON	1.5
	ON	OFF	2.5
	ON	ON	3
TAM7A0C36H31SD TAM7A0C36H31EA	OFF	OFF	3
	OFF	ON	2
	ON	OFF	2.5
	ON	ON	3.5
TAM7A0C42H31SD TAM7A0C42H31EA	OFF	OFF	3.5
	OFF	ON	2.5
	ON	OFF	3
	ON	ON	4
TAM7A0C48H41SD TAM7A0C48H41EA	OFF	OFF	4
	OFF	ON	3
	ON	OFF	3.5
	ON	ON	4.5**
TAM7B0C60H51SC TAM7B0C60H51EA	OFF	OFF	4.5**
	OFF	ON	3.5
	ON	OFF	4
	ON	ON	5

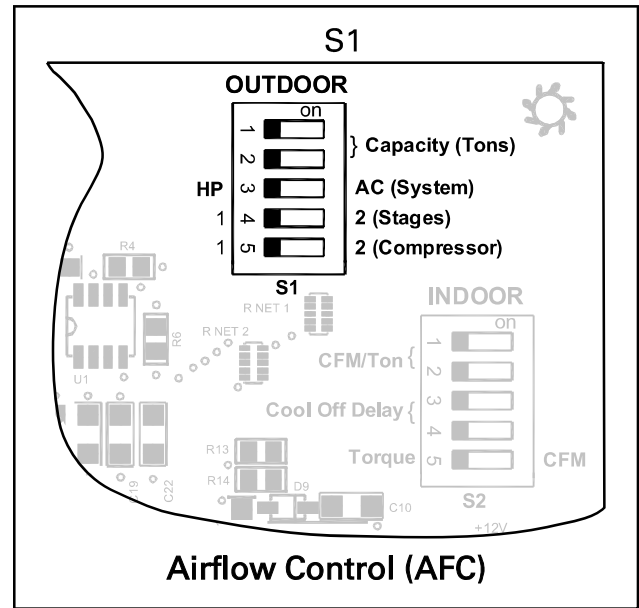
Note: The OD multiplier in conjunction with the CFM/TON can be used to adjust total airflow for your application.

Example: 4.5T x 370 CFM/TON = 1665 CFM

** Not an actual OD size

- Set the S1-3 dip switch for AC or HP.

OD TYPE	S1-3
HP	OFF
AC	ON



- Set the S1-4 dip switch for the number of stages on the outdoor unit.

OD STAGES	S1-4
1	OFF
2	ON

Note: Refer to the field wiring diagram of the Outdoor Unit to determine if the unit is 1 or 2 stage.

Note: This dip switch should be changed while the power is off or no demand is present. The power must be shut off and then re-applied in order for the AFC to recognize the change.

- Set the S1-5 dip switch for the number of compressors.

# COMPRESSORS	S1-5
1	OFF
2	ON

Dip Switches

Dip Switch S2

- Set the S2-1 and S2-2 dip switches for the Cooling and Heating CFM/Ton.

INDOOR MODEL	S2-1	S2-2	COOLING CFM/TON	HEATING CFM/TON
TAM7A0A24H21SD TAM7A0A24H21EA	OFF	OFF	400	430
	OFF	ON	360	390
	ON	OFF	380	410
	ON	ON	420	450
TAM7A0B30H21SD TAM7A0B30H21EA	OFF	OFF	400	430
	OFF	ON	360	390
	ON	OFF	380	410
	ON	ON	420	450
TAM7A0C36H31SD TAM7A0C36H31EA	OFF	OFF	370	420
	OFF	ON	350	400
	ON	OFF	390	440
	ON	ON	410	450
TAM7A0C42H31SD TAM7A0C42H31EA	OFF	OFF	370	400
	OFF	ON	330	360
	ON	OFF	350	380
	ON	ON	390	420
TAM7A0C48H41SD TAM7A0C48H41EA	OFF	OFF	350	400
	OFF	ON	330	380
	ON	OFF	370	420
	ON	ON	390	440
TAM7B0C60H51SC TAM7B0C60H51EA	OFF	OFF	370	400
	OFF	ON	380	410
	ON	OFF	390	420
	ON	ON	400	430

- Set the S2-3 and S2-4 dip switches for the Blower Off and Delay.

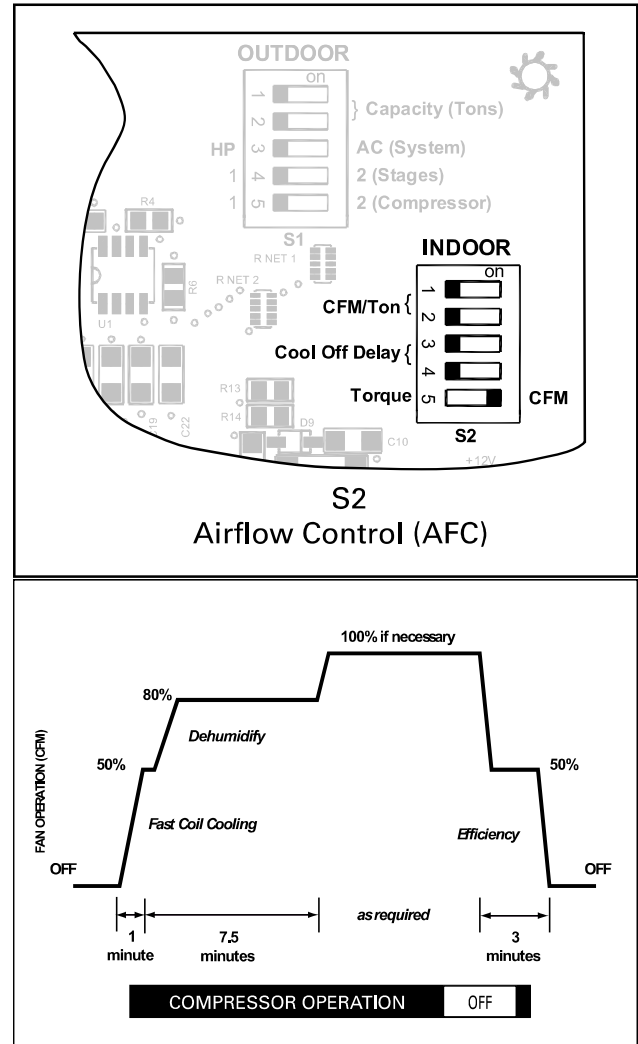
BLOWER OFF DELAY	S2-3	S2-4
1.5M @ 100%	OFF	OFF
NO DELAY	OFF	ON
3M @ 50%	ON	OFF
ENHANCED	ON	ON

Indoor Blower Timing

This ENHANCED MODE selection provides a ramping up and ramping down of the blower speed to provide improved comfort, quietness, and potential energy savings. The graph on the right shows the ramping process. Enhanced Mode works in cooling, HP, heating, Torque, and constant CFM modes.

- Set the S2-5 dip switch for the airflow mode.

AIRFLOW MODE	S2-5
*TORQUE	OFF
CONSTANT CFM	ON



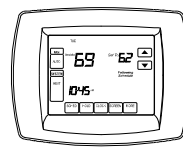
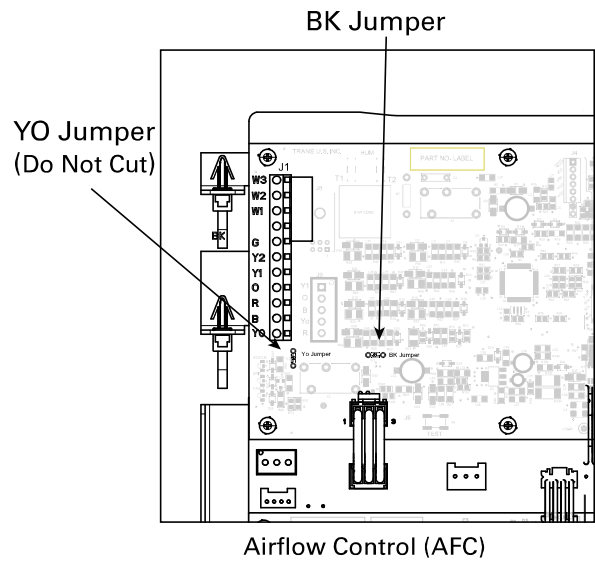
- All zoning applications MUST be set to Constant CFM mode.
- Torque mode is optional and is recommended for high static applications. High static conditions dramatically increase energy consumption with variable speed motors. Evaluate the blower performance table in relationship to the duct system to determine the benefit of each installation.
- For the 2 ton air handler, torque mode will reduce airflow when static is above approximately 0.3" water column.
- For the 2-1/2, 3, and 3-1/2 ton air handlers, torque mode will reduce airflow when static is above approximately 0.35" water column.
- For the 4 and 5 ton air handlers, torque mode will reduce airflow when static is above approximately 0.4" water column.

When selected, Torque Mode is used for cooling operation only.

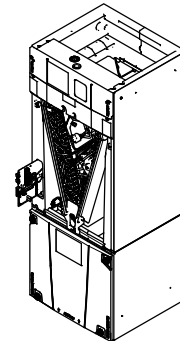
All heating modes default to constant CFM.

Indoor Blower Options

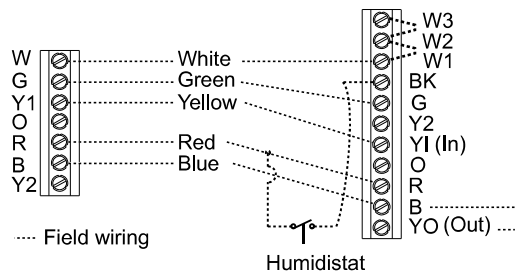
- Humidity Control for Cooling — This unit is capable of providing additional dehumidification during the cooling mode by reducing the airflow when humidity is above the set point. A conventional field supplied humidistat is required to control this function.
 1. Cut the BK jumper on the AFC.
 2. Using a field supplied humidistat, wire 24VAC Hot to one side of the humidistat. Wire the other side of the humidistat to the BK terminal on the low voltage terminal board. See wiring diagram.
- Pulse Width Modulation (PWM) — Typically used with A/ TCONT402 or zoning control.
 1. The BK jumper must be cut on the AFC.
 — Do Not cut the YO jumper.
- All zoning applications MUST be set to Constant CFM mode.



Comfort Control

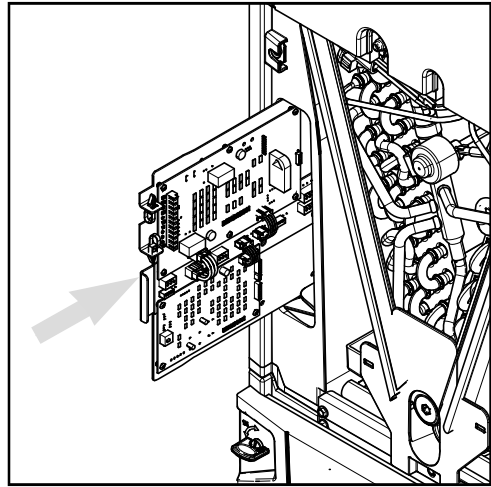


Air Handler

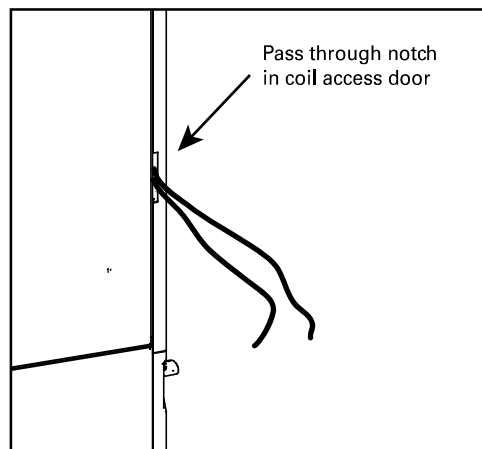
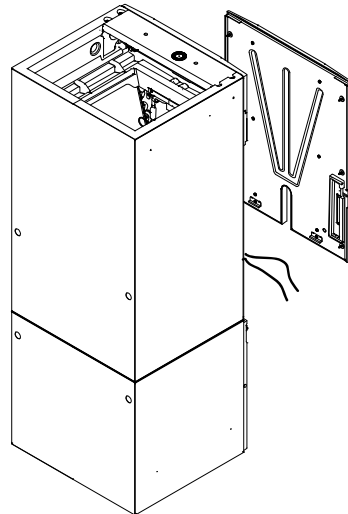


Control Plate Reinstall

1. Slide control plate assembly into the control pocket until fully seated. The control plate should be flush with the outer edge of the unit.



2. Replace coil panel making sure that the wires are located within the wire pass-through provided in the panel.

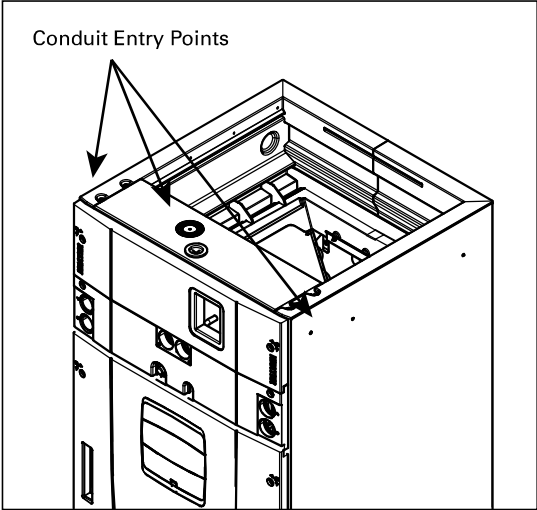
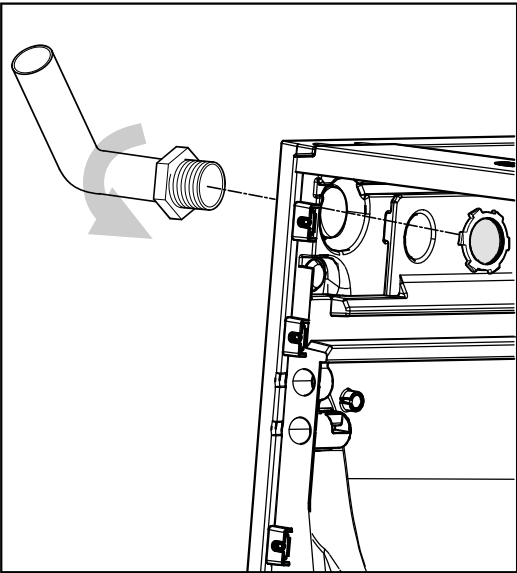


Electrical — High Voltage

Table 18. High Voltage Power Supply

The high voltage power supply must match the equipment nameplate. Power wiring, including ground wiring must comply with national, state, and local codes. Field wiring diagrams for supplementary electric heaters are shipped with the heaters.	⚠ WARNING LIVE ELECTRICAL COMPONENTS! Follow all electrical safety precautions when exposed to live electrical components. It may be necessary to work with live electrical components during installation, testing, servicing, and troubleshooting of this product.
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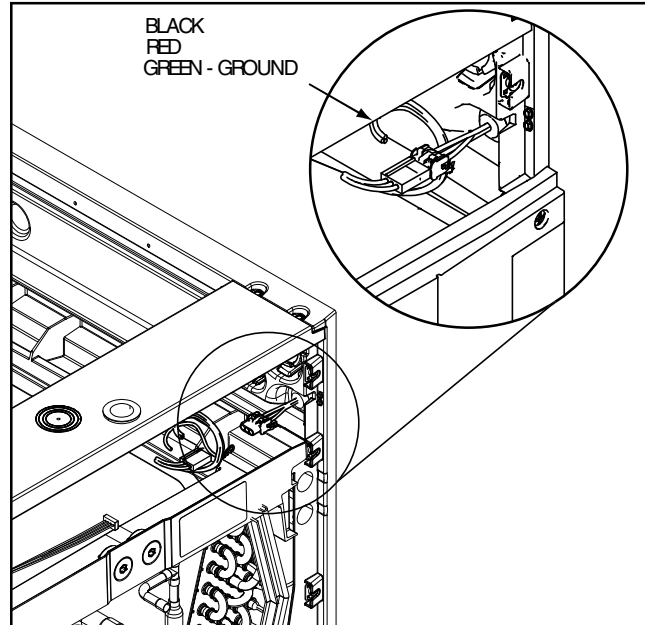
Table 19. Make Electrical Connections

1. Route High Voltage wiring to unit. 2. Select a conduit entry point. Drill a hole for the desired conduit size on units without a plug. A locating target is identified on these units. a. Select the entry point you will use to bring in your high voltage wiring. b. Remove plug from the entry point.	Conduit Entry Points 
3. Route conduit (if used) to the entry point and connect. a. Use one hand to secure the conduit nut from inside of the heater compartment. b. Connect a field supplied 3/4" or 1-1/2" conduit to conduit nut. Note: Reducing bushings may be required for your application.	

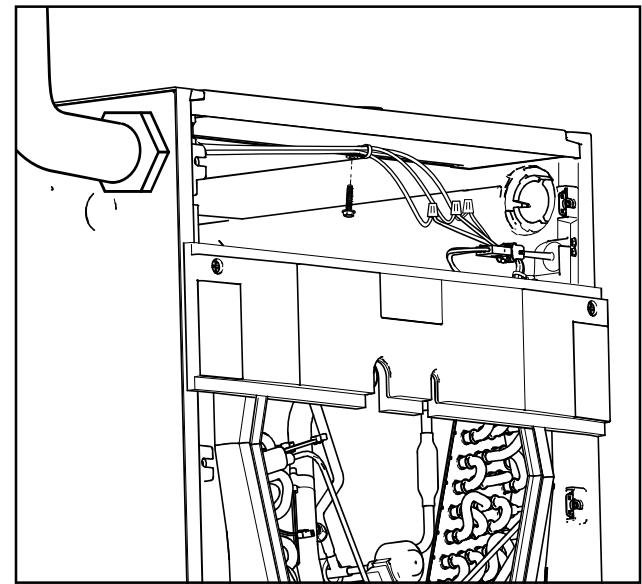
4. If an electric heater IS NOT being installed, remove the pigtail harness from the documentation pack and connect it to the plug on the inside of the Heater Compartment in the cabinet.

If an electric heater IS being installed, see the Installer's Guide shipped with the electric heater.

Connect L1, L2, and ground wiring to pigtail harness in Heater Compartment using wire nuts. The incoming ground wiring will mate up with the green wire shown in the illustration.

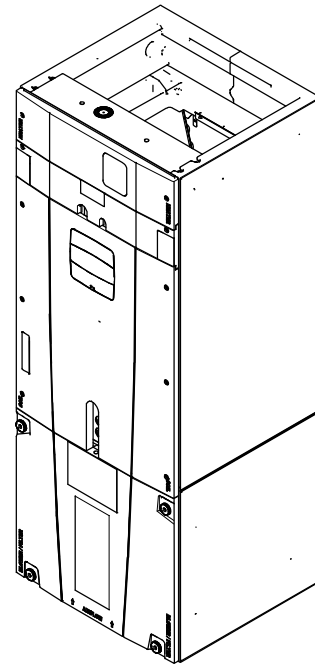


5. If the L1, L2, and ground wires enter the case from the left side, use a field supplied 1/2" — 5/8" maximum length screw and wire tie to hold the wires to the top center of the Heater Compartment.



6. Reinstall all panels before starting the air handler

Note: After replacing all panels, loosen the Line Set Panel screws approximately 1/4 — 1/2 turn. This will improve the seal between the Heater Panel and Line Set Panel.



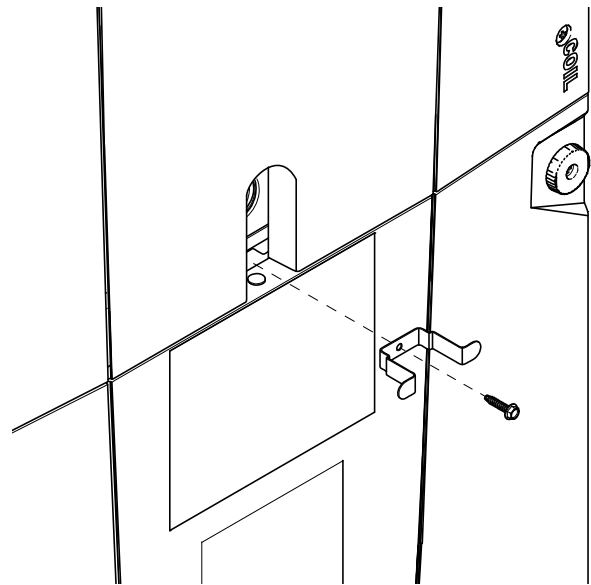
7. Remove screw and coil bracket from documentation packet.

8. Place the coil panel bracket into position and use screw to secure the coil panel bracket and seal plate to the support bar.

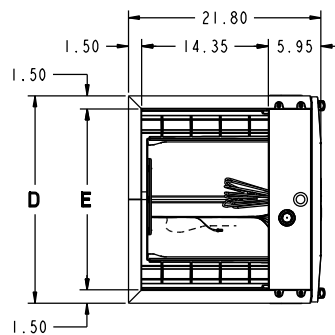
Important: The Coil Seal Plate and screw secure the coil in the center of the air handler. Failure to follow these steps can prevent the Coil Panel from being easily replaced on the unit.

Important: The Blower Panel may be removed if needed to help align the new screw with the seal plate and crossmember.

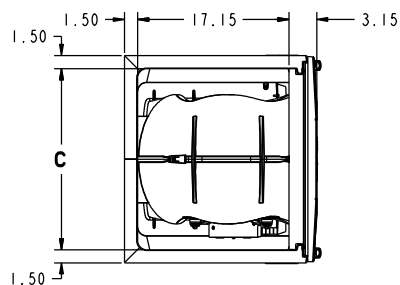
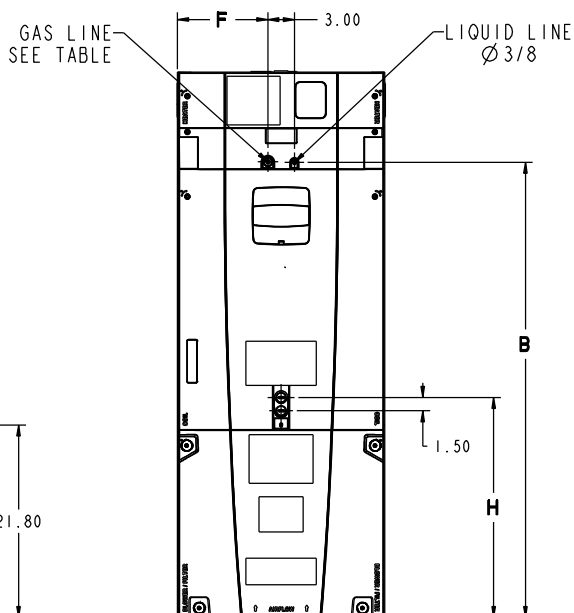
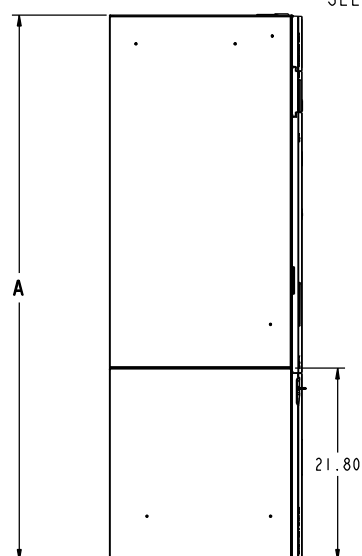
Important: For the 5 ton air handler model, tap 5 should not be used in the downflow or horizontal orientations. Using Tap 5 could result in water blowing off the coil.



TAM7 OUTLINE DRAWING



MINIMUM UNIT CLEARANCE TABLE		
	TO COMBUSTIBLE MATERIAL (REQUIRED)	SERVICE CLEARANCE (RECOMMENDED)
SIDES	0"	2"
FRONT	0"	21"
BACK	0"	0"
INLET DUCT	0"	
OUTLET DUCT	0"	



Model Number	A	B	C	D	E	F	H	FLOW CONTROL	GAS LINE BRAZE	LIQ LINE BRAZE
TAM7A0A24H21SD TAM7A0A24H21EA	49.9	39.6	14.5	17.5	14.5	7.3	24.4	EEV	3/4	3/8
TAM7A0B30H21SD TAM7A0B30H21EA	55.7	45.5	18.4	21.3	18.4	9.2	24.8	EEV	3/4	3/8
TAM7A0C36H31SD TAM7A0C36H31EA	56.9	46.7	20.5	23.5	20.5	10.3	24.2	EEV	7/8	3/8
TAM7A0C42H31SD TAM7A0C42H31EA	56.9	46.7	20.5	23.5	20.5	10.3	24.5	EEV	7/8	3/8
TAM7A0C48H41SD TAM7A0C48H41EA	61.7	51.5	20.5	23.5	20.5	10.3	24.9	EEV	7/8	3/8
TAM7B0C60H51SC TAM7B0C60H51EA	61.7	51.5	20.5	23.5	20.5	10.3	24.9	EEV	7/8	3/8

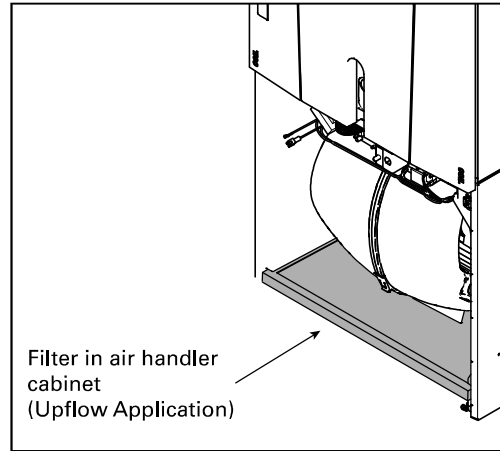
Filters

Table 20. Filter Considerations

- A filter must be installed within the system.
- A filter channel is provided in the unit at the bottom of the Blower/Filter compartment.
- For customer ease of filter maintenance, it is recommended that a properly sized remote filter grill(s) be installed for units that are difficult to access. Airflow should not exceed the maximum rated velocity of the filter being used.

Table 21. Filter Sizes

Cabinet Size *	A	B	C
Filter Size	16 x 20	20 x 20	22 x 20
* Cabinet size is indicated by the 7th digit in model number.			



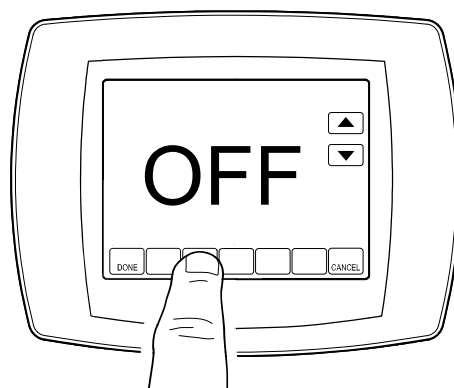
System Charge Adjustments

System Matched with:	Indoor Unit Model No.	Outdoor Unit Model No.	Subcooling
Single Compressor 2-Stage HP	TAM7A0B30H21SD TAM7A0B30H21EA	4A6H6024E/G, 4TWX6024E/G 4A6H7024, 4TWX8024	9°
	TAM7A0C36H31SD TAM7A0C36H31EA	4A6H6036E/G, 4TWX6036E/G 4A6H7036, 4TWX8036	10°
	TAM7A0C48H41SD TAM7A0C48H41EA	4A6H6048E/G, 4TWX6048E/G 4A6H7048, 4TWX8048	8°
Single Compressor 2-Stage AC	TAM7A0B30H21SD TAM7A0B30H21EA	4A7A6024E/G, 4TTX6024E/G 4A7A7024, 4TTX8024	8°
	TAM7A0C36H31SD TAM7A0C36H31EA	4A7A6036E/G, 4TTX6036E/G 4A7A7036, 4TTX8036	8°
	TAM7A0C48H41SD TAM7A0C48H41EA	4A7A6048E/G, 4TTX6048E/G 4A7A7048, 4TTX8048	8°
Two Compressor 2-Stage HP	TAM7A0B30H21SD TAM7A0B30H21EA	4TWZ0024A, 4A6Z0024A	9°
	TAM7A0C36H31SD TAM7A0C36H31EA	4TWZ0036A/B, 4A6Z0036A/B	10°
	TAM7A0C48H41SD TAM7A0C48H41EA	4TWZ0048A/B, 4A6Z0048A/B	12°
	TAM7B0C60H51SD TAM7B0C60H51EA	4TWZ0060A, 4A6Z0060A	12°
Two Compressor 2-Stage HP	TAM7A0B30H21SD TAM7A0B30H21EA	4TTZ0036A/B, 4A7Z0024A	9°
	TAM7A0C36H31SD TAM7A0C36H31EA	4ATTZ0036A/B, 4A7Z0036A/B	11°
	TAM7A0C48H41SD TAM7A0C48H41EA	4TTZ0048A/B, 4A7Z0048A/B	13°
NOTES: 1. Variable Speed outdoor units must be charged per the outdoor unit instructions. 2. All other matches must be charged per the nameplate charging instructions.			

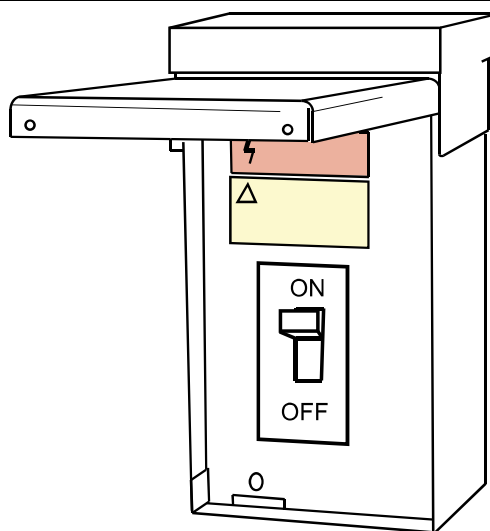
System Start Up

1. Make sure all panels are securely in place and that all wiring has been properly dressed and secured.

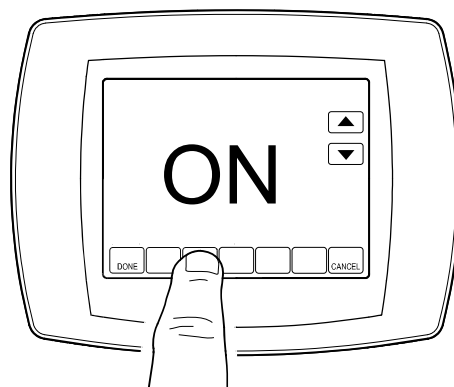
2. Set the system thermostat to OFF.



3. Turn on electrical power disconnect(s) to apply power to the indoor and outdoor units.



4. Set the system thermostat to ON.



TAM7 Sequence of Operation

Abbreviations

- AFC = Airflow Control
- EVC = Expansion Valve Control
- EEV = Electronic Expansion Valve
- EHC = Electric Heat Control
- HHC= Hydronic Heat Control
- Y1 = Y Signal from the comfort control
- YO = Y Signal out of the AFC from the comfort control
- Y2 = Y2 Signal from comfort control

This unit has a unique feature that receives 24VAC analog inputs from a conventional comfort control and then converts these inputs to serial communicating signals between the different controls within the air handler.

The installing and servicing technician should have an understanding of the sequence of operation to be able to properly setup and diagnose functions of the air handler.

The dip switch settings on the control boards must be properly set to obtain the correct sequence of operation. Reference the air handler and electric heat wiring diagrams and dip switch configuration chart for additional information.

See unit, electric heat, and field wiring diagrams for additional information.

Continuous Fan

Important: *If the indoor air exceeds 60% relative humidity or simply feels uncomfortably humid, it is recommended that the indoor fan only be used in the AUTO mode.*

1. R-G contacts close on comfort control sending 24VAC to the G terminal of the AFC.
2. The AFC micro-processor sends a command to the serial communicating blower motor to run at 50%.

See airflow Dip Switch settings for additional info.

Heat pump OD Single Stage (heating mode)

1. R-Y contacts close on the comfort control sending 24VAC to Y1 terminal on the AFC.
2. Normally closed contacts on the AFC will pass 24VAC to the YO terminal providing power to the outdoor unit control circuit.

3. R-G contacts close on comfort control sending 24VAC to the G terminal of the AFC.
4. The AFC micro-processor sends a command to the serial communicating blower motor to run at 100%.
5. The HUM contacts close.

Heat pump OD Two Stage (heating mode)

1. R-Y contacts close on the comfort control sending 24VAC to Y1 terminal on the AFC.
2. Normally closed contacts on the AFC will pass 24VAC to the YO terminal providing power to the outdoor unit 1st stage control circuit.
3. R-G contacts close on comfort control sending 24VAC to the G terminal of the AFC.
4. The AFC micro-processor sends a command to the serial communicating blower motor to run at 1st stage heating cfm.
5. The HUM contacts close.
6. R-Y2 contacts close on the comfort control sending 24VAC to Y2 terminal on the AFC.
7. The Y2 signal is also passed through to provide power to the outdoor unit 2nd stage control circuit.
8. The blower motor will now run at 100% of the heating cfm.

OD Single Stage (cooling mode)

1. R-Y contacts close on the comfort control sending 24VAC to Y1 terminal on the AFC.
2. Normally closed contacts on the AFC will pass 24VAC to the YO terminal providing power to the outdoor unit control circuit.
 - a. Heat pump OD > R-O contacts close on the comfort control sending 24VAC to the O terminal of the AFC. The AFC communicates to the EVC that the system is in cooling mode. The O signal is passed through to the OD unit.
 - b. Non heat pump OD > OD dip switch configuration will communicate to the EVC to control superheat on a Y only signal.
3. R-G contacts close on comfort control sending 24VAC to the G terminal of the AFC.
4. The AFC micro-processor sends a command to the serial communicating blower motor to run at 100% of the cooling cfm.
5. The EVC will receive input from the two temperature sensors and start to control superheat.

OD Two Stage (cooling mode)

1. R-Y1 contacts close on the comfort control sending 24VAC to Y1 terminal on the AFC. The AFC communicates to the EVC to control superheat for 1st stage.
2. Normally closed contacts on the AFC will pass 24VAC to the YO terminal providing power to the outdoor unit 1st stage control circuit.
 - a. **Heat pump** OD > R-O contacts close on the comfort control sending 24VAC to the O terminal of the AFC. The AFC communicates to the EVC that the system is in cooling mode. The O signal is passed through to the OD unit.
 - b. **Non-Heat pump** OD > OD dip switch configuration will communicate to the EVC to control superheat on a Y only signal.
3. R-G contacts close on comfort control sending 24VAC to the G terminal of the AFC.
4. The AFC micro-processor sends a command to the serial communicating blower motor to run at the 1st stage cooling cfm. (1st stage cooling cfm is a multiplier of 2nd stage cfm)
5. The EVC will receive input from the two temperature sensors and start to control 1st stage superheat.
6. R-Y2 contacts close on the comfort control sending 24VAC to Y2 terminal on the AFC. The AFC communicates to the EVC to control superheat for 2nd stage.
7. The Y2 signal is passed through to provide power to the outdoor unit 2nd stage control circuit.
8. The blower motor will now run at 100% of the cooling cfm.

Electric Heat

1. R-W contacts close on the comfort control sending 24VAC to W1 of the AFC.
2. R-G contacts close on the comfort control sending 24VAC to G of the AFC.
3. The AFC communicates to the EHC that 1st stage electric heat is being called upon.
4. The EHC determines the number of elements that are used for 1st stage and sends a message to the AFC for that correct cfm. (The EHC determines the amount of heat per stage by either factory programming or by the kw jumper position)
5. The AFC micro-processor sends a command to the serial communicating blower motor to run and close the blower interlock relay on the EHC. The blower motor will now run at the W1 electric heat cfm.
6. On subsequent calls for W2 and/or W3, the EHC will communicate to the AFC the required airflow request and energize the additional relays.

Note: The EHC has "lead-lag" logic built in that energizes the electric heat relays based upon cycle counts.

For example: BAYEV**15 – The first time W1 only is energized; the K1 relay would close and energize the "A" heater. The second time W1 only is energized; the K2 relay would close and energize the "B" heater. The third time W1 only is energized; the K3 relay would close and energize the "C" heater.

Defrost

1. The OD unit will initiate defrost and send 24VAC to the O terminal of the AFC.
2. The AFC will communicate to the EVC that the OD is in defrost and the EVC will start to maintain the correct superheat.
3. X2 from the OD will send 24VAC to W1 of the AFC.
4. The AFC communicates to the EHC that 1st stage electric heat is being called upon.
5. The EHC determines the number of elements that are used for 1st stage and sends a message to the AFC for that correct cfm.

Optional Condensate Switch

1. An optional OEM condensate switch can be installed within the unit. This switch is only available through the National Distribution Center or Global Parts.
2. Switch contacts are normally open and close when water level rises. The closed switch will interrupt current flow to the YO terminal and de-energize the OD unit.
3. Switch is only operational during cooling mode. Condensate overflow is not operational during heating or defrost modes.

Standard aftermarket condensate switches cannot be used within the unit but can be installed on exterior of the unit. The 3rd party condensate overflow switch must be wired between Y of the thermostat and Y1 of the AFC control.

Freeze Protection

1. The EVC control has the ability to sense when the coil is beginning to ice. When this event occurs, the contacts to the YO circuit will open and de-energize the OD unit.
2. The indoor blower motor will continue to run to aid in defrosting the coil. After 5 minutes, the YO contacts will close and cooling operation will begin again.

Blower Delays

1. Blower delays can be set to enhance system efficiency. See S2 dip switch settings table.

Checkout Procedures

The final phase of the installation is the system Checkout Procedures. The following list represents the most common items covered in a Checkout Procedure. Confirm all requirements in this document have been met.

☐ All wiring connections are tight and properly secured. ☐ Voltage and running current are within limits. ☐ All refrigerant lines (internal and external to equipment) are isolated, secure, and not in direct contact with each other or structure. ☐ All braze connections have been checked for leaks. A vacuum of 350 microns provides confirmation that the refrigeration system is leak free and dry. ☐ Final unit inspection to confirm factory tubing has not shifted during shipment. Adjust tubing if necessary so tubes do not rub against each other or any component when unit runs. ☐ Ductwork is sealed and insulated. ☐ All drain lines are clear with joints properly sealed. Pour water into drain pan to confirm proper drainage.	☐ Supply registers and return grilles are open, unobstructed, and air filter is installed. ☐ Indoor blower and outdoor fan are operating smoothly and without obstruction. ☐ Indoor blower motor set on correct speed setting to deliver required CFM. ☐ Cover panels are in place and properly tightened. ☐ For gas heating systems, manifold pressure has been checked and all gas line connections are tight and leak free. ☐ For gas heating systems, flue gas is properly vented. ☐ System functions safely and properly in all modes. ☐ Owner has been instructed on use of system and given manual.
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The manufacturer optimizes the performance of homes and buildings around the world. A business of Ingersoll Rand, the leader in creating and sustaining safe, comfortable and energy efficient environments, the manufacturer offers a broad portfolio of advanced controls and HVAC systems, comprehensive building services, and parts. For more information, visit www.IRCO.com.

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